

Behavioral Lock-In: Aggregate Implications of Reference Dependence in the Housing Market

Cristian Badarinza, Tarun Ramadorai,
Juhana Siljander, Jagdish Tripathy*

This paper is evolving. For the latest version, please [click here](#).

August 22, 2024

Abstract

Households' attachment to nominal anchors in the housing market creates aggregate nominal rigidities, with material economic consequences. A new statistic, the prevalence of "paper losses" in the stock of residential properties, captures cross-regional variation in important housing market quantities; the share of properties facing paper losses explains housing transactions volumes more effectively than price growth rates. To rationalize these patterns and study housing-market fiscal policy, we develop and structurally estimate a dynamic search and matching model with reference-dependent agents, rich heterogeneity, and realistic financial constraints. Behavioral frictions dampen the tax elasticity of property prices, and increase the revenue-maximizing level of property taxes.

*Badarinza: NUS Business School, Email: cristian.badarinza@nus.edu.sg. Ramadorai (corresponding author): Imperial College London and CEPR, Email: t.ramadorai@imperial.ac.uk. Siljander: Imperial College London, Email: j.siljander@imperial.ac.uk. Tripathy: Bank of England, Email: jagdish.tripathy@bankofengland.co.uk. Badarinza and Tripathy were visiting researchers at Imperial College London during the production of this draft. We thank Ed Burgess, Tim Bannister, Natalie Jakomis, Leanne Pienaar, Amit Tiwari, and Oscar Walet at Rightmove for data access and numerous useful conversations, and seminar participants at the London School of Economics Macro Workshop, the Imperial-LSE-UCL-IFS-CEPR Micro-Macro Household Finance Seminar, the London Behavioural Finance Workshop, Princeton University, Emory University, and Oxford University for comments. Any views expressed in this article are not meant to represent the views of the Bank of England, members of its Financial or Monetary Policy Committee, or the Financial Conduct Authority.

1 Introduction

Nominal rigidities affect the transmission of both monetary and fiscal policy, and the literature has identified many underlying sources of these rigidities in systemically important markets. These include firm-level adjustment costs (Barro, 1972), sticky wage contracting (Calvo, 1983), imperfect information (Mankiw and Reis, 2006), and more recently, the inattentive and inertial behavior of economic agents (Reis, 2006a,b; Abel et al., 2013; Andersen et al., 2020). In this paper, we introduce a novel and important source of aggregate nominal rigidities to the study of housing market fluctuations, which comes from the well-known behavioral attachment of agents to a nominal anchor.¹

We document the importance of this channel for aggregate housing market outcomes both theoretically and empirically. Our empirical work harnesses rich and granular administrative data on housing transactions, property listings and mortgage borrowing in the UK over the period between 2010–2022. Using these data, we show that a single statistic, the prevalence of “paper losses” in the stock of residential properties (henceforth the “loss share”), effectively explains cross-regional variation in a range of housing market outcomes. The loss share summarizes the distribution of nominal anchors in the population of homeowners; as we demonstrate in our theory, it serves as a sufficient statistic that aids optimal policy calculations (Chetty, 2009).

In our theoretical work, we build a dynamic search and matching model of the housing market with reference-dependent and loss-averse agents, featuring rich heterogeneity and realistic financial constraints. We structurally estimate the parameters of the model using a set of micro moments that capture well-studied household behavior in the housing market (Genesove and Mayer, 2001; Bracke and Tenreyro, 2021; Andersen et al., 2022). The resulting calibrated model is well able to generate the patterns documented in the data, even though these aggregate moments are untargeted in estimation.

We use the model to analyze the ongoing debate on how fiscal policy affects the

¹As predicted by Kahneman and Tversky (1979), people define utility relative to a subjective reference point, and weight losses more than gains. Careful experiments and field work have documented the presence and importance of such preferences across a wide range of settings. This holds in terms of daily life activities (Allen et al., 2017), for high-stakes decisions that involve job search (DellaVigna et al., 2017), the filing of income taxes (Rees-Jones, 2018), financial investment (Ingersoll and Jin, 2013), and housing (Genesove and Han, 2012; Bracke and Tenreyro, 2021; Andersen et al., 2022).

housing market.² We distinguish between two types of taxes: an ongoing property tax, which applies annually as a proportion of the property value (“council tax” in the U.K.), and an ownership transfer tax levied on the buyer of a property (“stamp duty land tax” in the U.K.). Both types of taxes are associated with a Laffer-type hump-shaped relationship between the level of the tax and the total government revenue raised, in the rational baseline model as well as the model with behavioral agents.

When agents are reference-dependent, equilibrium prices are less sensitive to the prevailing tax regime. This is because behavioral sellers trade off any tax loss or benefit against the probability of realizing an additional nominal loss or gain on the sale of the property. As nominal anchoring decreases the elasticity of prices and increases the response of volumes, both the shape of the Laffer curve and the optimal revenue-maximizing tax rate depend on the prevailing loss share in a given location at any given point in time. We validate this insight, documenting that the degree of bunching of transactions prices around historical notches in the U.K. stamp duty tax schedule depends on the size of the loss share. This exercise reinforces the usefulness of the loss share as a sufficient statistic for evaluating tax policy changes in the housing market.

We first verify a long-known fact about housing markets in our data, namely, the strong positive comovement between changes in nominal house prices and housing sale transaction volume. This is evident across U.K. local authority districts; we verify that the same behavior shows up in data from U.S. states. This similarity in the behavior of the two housing markets is also evident in the magnitude of comovement, despite their very different mortgage market structures.³

We next show that the loss share in a given location at a given point in time accounts for a large part of the comovement between changes in prices and transactions volumes. The coefficient of price growth rates in a regression explaining transaction volumes decreases, becoming statistically insignificantly different from zero in most specifications, when we introduce the loss share into the regression. This result is

²See, for example, “Rent controls will not fix Britain’s housing crisis”, *Financial Times*, 27 May 2024, for a summary of the current debate on the reform of the council tax system in the U.K.

³The U.S. market is dominated by fixed-rate mortgages, and there is recent evidence (Fonseca and Liu, 2023) that “mortgage lock-in” affects homeowner mobility. In contrast, the U.K. market is dominated by legally portable short-duration adjustable-rate mortgages with a modal fixation period of two years, meaning that there are strong refinancing incentives at such short horizons (Fisher et al., 2023), and potentially very little mortgage lock-in.

robust to using a hedonic model to compute price growth rates, or to calculating price changes, dollar volumes, and loss shares from aggregate price indices which rely only on publicly available transaction data computed using repeat sales methods.⁴ Intuitively, when many owners are reluctant to realize nominal-loss-inducing sales, house price declines are accompanied by substantial declines in housing market volume. We also show that a high share of paper losses in a region predicts high volatility of volumes, with no corresponding relationship for price volatility. This finding is consistent with the well-known observation that housing market crashes are first visible in market liquidity, and then only gradually reflected in prices.

We build a dynamic equilibrium search and matching model of the housing market to rationalize these patterns in the data. The model features agents that are reference dependent and loss averse,⁵ but apart from these behavioral preferences, they are forward-looking, rational, and fully informed about aggregate quantities, shocks, and the distributions of reference points and mortgage balances in the population. Agents face realistic mortgage down-payment constraints, which allows for a separate and distinct role for the effect of financial leverage on house selling decisions, first identified by [Stein \(1995\)](#).

Potential sellers (homeowners) in the model receive idiosyncratic mobility (or productivity) shocks that motivate them to sell. They trade off these mobility payoffs optimally against the nominal gains and losses that they make from selling their houses at prices above or below a pre-determined reference point, and against penalties for falling below a pre-determined down-payment level of home equity.⁶ They evaluate these tradeoffs dynamically in an infinite horizon setup, optimally solving for whether or not to list in each period, and choosing a pricing strategy conditional on listing. In the event that they sell, they return to the market as potential buyers in the next period, and take this into account when optimizing.⁷

Buyers receive idiosyncratic taste shocks when they are randomly matched with

⁴Repeat sales approaches are widely used in the housing market literature, dating back at least to [Bailey, Muth and Nourse \(1963\)](#) and [Shiller \(1991\)](#)

⁵Technically, agents realize utility à la [Barberis and Xiong \(2012\)](#) from nominal (log) gains and losses relative to their historical nominal reference point, in addition to the “standard” log utility that they derive from the final sale price.

⁶Outstanding mortgage amounts evolve according to a pre-specified amortization schedule. With heterogeneous holding periods, this generates rich heterogeneity in homeowners’ mortgage balances, with a methodological resemblance to the quantitative overlapping generations models that have a long history in the public finance literature ([Auerbach and Kotlikoff, 1987](#)).

⁷In a downturn, sellers rationally anticipate the fact that selling their property at a low price might subsequently enable them to purchase another property at a similarly low price.

properties, and also optimize dynamically, trading off the probability of a successful transaction today against waiting for more promising outcomes in the future. When buyers in the model purchase properties, they become homeowners/prospective sellers, and their purchase prices are encoded as their reference prices for future property sale decisions. The distribution of reference prices in the economy thus varies endogenously with the realized set of transactions. We solve for the stationary distribution of these reference prices as an equilibrium quantity, which means that the model endogenizes the prevalence of “paper losses” in the economy.

Consistent with the implications of [Krusell and Smith \(1998\)](#), [Lucas \(2003\)](#) and [Ahn et al. \(2018\)](#), the heterogeneity of individual state variables matters for aggregate equilibrium outcomes, because of agents’ reference dependence, loss aversion, and occasionally binding mortgage constraints. More specifically, buyers’ and sellers’ nonlinear decision rules generate aggregate externalities through their impact on market tightness (which determines the rate at which buyers and sellers meet one another), and the probability of sale conditional on a meeting (which reflects the full distribution of seller list prices).

We calibrate the parameters of the model to match a set of well-studied micro-level moments which capture buyer and seller behavior in the housing market. One novel aspect of our data is that in addition to the listing behavior of sellers, we also observe the search activity of potential buyers (the data track online search as well as requests for offline/physical meetings). This allows us to parsimoniously quantify the magnitude of the search frictions present at any given time, by measuring the extent of market tightness, i.e., the ratio of prospective buyers to sellers present in the housing market data.

At the aggregate level, the model shows that different locations should exhibit different responses to shocks, depending on the mass of local homeowners that face nominal loss-making transactions. More specifically, the elasticity of housing transaction volumes to shocks affecting the utility of homeownership depends on the distribution of potential nominal gains and losses in the population of homeowners. Consistent with the data, the model-implied volume response to price-relevant shocks is also asymmetric, i.e., negative shocks to housing value generate a greater transaction volume impact than positive shocks.

The model also shows that the elasticity of listing volumes to price changes (i.e., the effects of prices on the “extensive margin” decision of whether homeowners list properties up for sale at all) is very small. This is because effects on the decision

to list are partially absorbed by the optimal pricing strategy conditional on listing, i.e., the intensive margin of the listing decision. Put differently, the model predicts that it is almost always optimal for homeowners to hang out a “speculative” listing rather than to avoid listing in the face of a nominal loss. This insight is consistent with the data, where we show that sellers list their properties for sale at roughly similar rates in “hot” and “cold” markets (Ngai and Tenreyro, 2014).

An important implication of the model is that sellers that face the possibility of realizing nominal losses can become “behaviorally locked-in,” eschewing potentially valuable moving opportunities. A positive nominal change in property valuations generated by a higher rate of inflation can therefore lead homeowners to more readily accept moving opportunities, which supports the role of “location as an asset,” and speaks to the potentially positive role that inflation can play in a market with nominally-anchored agents.

As discussed earlier, standard optimal policy calculations can differ from canonical theory in the presence of behavioral agents. Our calibrated model is well equipped to analyse how optimal tax policy design should adjust in response to behavioral frictions in the housing market; and we conclude our analysis by assessing the consequences of different types of property taxation, both in terms of outcomes in the housing market as well as the impact on government revenue.

Our paper is related to the recent and growing literature that seeks to explore the role of behavioral frictions for macroeconomic and financial outcomes in a heterogeneous-agent setup (Krusell and Smith, 1998; Ahn et al., 2018), and the impact of behavioral frictions on optimal policy (O’Donoghue and Rabin, 2006; Mullainathan et al., 2012; Chetty, 2015; DellaVigna et al., 2017; Taubinsky and Rees-Jones, 2018; Rees-Jones, 2018; Allcott et al., 2019). As in many public finance applications, our empirical identification approach relies on a bunching estimator (Kleven, 2016; Allen et al., 2017; Rees-Jones, 2018; Anagol et al., 2022).

Our work emphasizes the asymmetric impact of shocks across different segments of the population (Ahn et al., 2018; Huckfeldt, 2022), and the persistence of price dispersion (Burdett and Judd, 1983). We propose an equilibrium model of the housing market that is in the tradition of standard search and matching frameworks (Diamond, 1984; Merlo and Ortalo-Magne, 2004; Diaz and Jerez, 2013; Han and Strange, 2015; Allen et al., 2019; Piazzesi et al., 2020), explicitly incorporating the endogenous determination of liquidity (Wheaton, 1990; Krainer, 2001; Ngai and Tenreyro, 2014; Guren, 2018; Anenberg and Ringo, 2022), the sequential nature of buying and

selling decisions (Anenberg and Bayer, 2020; Moen et al., 2021; Grindaker et al., 2021), and the role of financial frictions (Stein, 1995; Fisher et al., 2023; Fonseca and Liu, 2023).

The paper is organized as follows. Section 2 describes the sources that we employ to document both micro and aggregate patterns in the data. Section 3 describes the stylized aggregate facts from the U.K. market that motivate the development of the model. Section 4 introduces our dynamic search and matching model of the housing market. Section 5 discusses the calibration and structural estimation of the model, and discusses how the model fits (untargeted) patterns in aggregate data. Section 6 considers the implications of the model for housing market tax policy, documents evidence in the data on historical tax policy responses, and outlines other implications of reference dependence for economic policy. Section 7 concludes.

2 Data

Our analysis uses four main data sets: address-level information on the U.K.’s residential property stock from the U.K. Ordnance Survey, the universe of residential property transactions in England and Wales from the HM Land Registry, listings for properties available for sale and corresponding online search information from Rightmove.com, and loan-level data on the outstanding stock of mortgage contracts from the Bank of England.

2.1 Residential property stock and transactions

2.1.1 Property stock

The U.K. residential property stock comprises 27.3 million postal residential addresses, recorded in the Royal Mail Postcode Address File, released by the Ordnance Survey. We use these address data in combination with house listings and sales data (described below) to compute the fraction of the housing stock that is turned over.⁸

⁸We implement our analysis at different levels of geographical aggregation. This is driven by the availability of data and suitability for the specific analysis being conducted. At the very top level of aggregation, we cover two nations (England and Wales), which together account for 10 International Territorial Level 1 (ITL1) and 35 Level 2 (ITL2) regions. These regions are further disaggregated into 339 local authorities (LAD), 7862 electoral wards, and 1.49 million residential postcodes. Because each postcode includes an average of 18.4 residential addresses, it is a precise indicator of the location of housing units. The online appendix provides an overview of the composition and

2.1.2 Property transactions

We obtain the universe of residential property transactions in England and Wales from HM Land Registry. The data include details on the address of the unit (including the property postcode), the date when the contractual sale agreement was signed, and the price recorded in the contract. In total, 28 million transactions were recorded between January 1995 and December 2022, corresponding to a turnover rate of the housing stock of around 2.5% for every 6 months.

For comparison purposes, we also acquire publicly available U.S. data on residential property sales from Zillow. From this source, we construct a panel of median prices and transaction volumes aggregated at the U.S. state level for the monthly frequency between January 2010 and December 2022.

2.2 Listings for sale

We acquire listing information for individual properties from Rightmove.com, the largest online portal for house listings in the UK. The listings data set tracks all listings on the portal that resulted in a successful transaction, as well as those that were withdrawn and archived. We merge these listings with the transactions registry. The merge rate is high—for example, 77% of Land Registry transactions in 2022 match with a corresponding preceding listing—consistent with the high market share of Rightmove.com and the predominant tendency of sale properties in the U.K. to be first listed online.

For each property, the data contain information on the exact address, the date of the initial listing, the listing price, and a comprehensive set of hedonic characteristics which include the number of bedrooms and bathrooms, the floor space area, the building type (detached and terraced houses, vs. apartments), and indicator variables for the construction type, whether the property is a development property, retirement home, or classified as affordable housing. The data cover 21 million listings between January 2010 and December 2022.

To merge listings with final transactions, we filter both data sets by restricting the geographical coverage to England and Wales and removing potentially misreported address units (those with either several transactions or several listings associated with a single address on the same day). We implement two different data merges using

filtering of the data.

postcode-address pairs to identify properties in both data sets uniquely: (i) we match listings to a preceding transaction, which allows us to estimate the nominal gain associated with the property; and (ii) we match transactions with preceding listings to facilitate estimation of a hedonic pricing model using the hedonic characteristics in the Rightmove listings data. For the hedonic model, we constrain to the set of properties that are transacted within 365 days of the first listing to avoid potential changes in hedonic characteristics from potentially “stale” listings information. We also remove auction properties, transactions that are completed within 30 days of the first property listing for sale, and properties for which we do not have sufficient hedonic information in the data. These filters leave us with a final data set of 3.9 million transactions on which we estimate the hedonic model.

2.3 Search and matching

Our data also record online search activity for all Rightmove.com listings observed between January 2019 and December 2022. We distinguish between two types of engagements between an online user and a property listing.⁹ The first, which we denote as a “search visit,” is an instance of the user clicking the “detail view” hyperlink on a property listing. This provides the full set of property characteristics to the user (e.g., the floor area) and qualitative descriptions of features that the seller/landlord deems relevant. The second layer of engagement, which we denote as a “meeting”, is when an online user enters a request for direct contact with the owner of the property or the real estate agency that manages the listing, to organize a physical viewing of the property. When we compute market tightness, i.e., the number of buyers divided by the number of listings, we use the number of initiated physical meetings as our measure of the number of potential buyers interested in each property.

2.4 Mortgage loans

We acquire information about the stock of mortgages issued in the U.K. from the Financial Conduct Authority’s PSD007 data set. These data cover the period between

⁹Each “online user” in our setting corresponds to a unique IP address tagged with a fully anonymized (hashed) user identifier. We filter the data to remove online users with only one search visit over the entire sample period and those who search over more than 1,000 listings, as these are likely bots.

2015 and 2022 and are available in half-yearly snapshots.

To obtain information about the outstanding mortgage balance at the time of listing for eventually transacted listings, we identify individual properties using the postcode, the previous transacted value and the mortgage contract account opening date. The combination of these three variables is a near-unique identifier of mortgages in our sample.

We approximate the current home equity position for each listing based on the outstanding balance from the half-yearly snapshot closest to and prior to the transaction date. In some cases, the outstanding balance from the snapshot closest to the transaction date is reported as 0 as a result of account closure; in such cases, we rely on the balances from the preceding snapshot.

In order to preserve privacy restrictions across the two data sets, we create binned versions of the home equity position of each mortgage, which are based on average values in bins of 10 observations created from the listings data. The online appendix describes this procedure in detail.

We also merge observations across mortgage-stock snapshots to estimate the average mortgage amortization schedule. To effect this merge, we track mortgages across stock snapshots based on the postcode and borrower date of birth (the latter variable also allows us to uniquely identify a borrower-property combination).

We rely on a sub-sample of mortgages originating in the first half of 2015 to construct an implied amortization schedule. This sample comprises 70,000 observations with an average origination loan size of £157,500, an initial interest rate of 2.95%, and a monthly payment of £776. Figure A.8 shows the corresponding estimated amortization schedule. On average, mortgages originating in 2015H1 amortize 1.85 pp of the loan balance every six months and are amortized by a cumulative 30.14 pp by 2022H2.

2.5 Counterfactual distributions: Hedonic model

To compute a counterfactual distribution of transaction prices, we compute the estimated hedonic value of each property to estimate an adjusted-for-characteristics price for each property. For the part of the sample for which we have high-quality measures of hedonic characteristics, we estimate a standard hedonic pricing model,

which predicts the log of the sale price P_{it} of all sold properties i in each year t :

$$\ln(P_{i,t}) = \zeta_w + \xi_{l,t} + \psi_{r,m} + \beta_{\mathbf{x}}' \mathbf{X}_{i,t} + \varepsilon_{i,t}, \quad (1)$$

where ζ_w are electoral ward fixed effects, $\xi_{l,t}$ are local authority district $\times$ year fixed effects, $\psi_{r,m}$ are region $\times$ month fixed effects, and $\mathbf{X}_{i,t}$ is a vector of time-varying property characteristics, namely, a second order polynomial of floor area (m^2) augmented with a logarithmic term, number of bedrooms and bathrooms, and dummy variables for property type, whether the property is a development property, retirement home, or affordable housing property. The baseline model has strong explanatory power with an R^2 of 0.87.¹⁰

2.6 “Paper losses” and home equity

Panel A of Figure A.10 shows the distribution of nominal “paper” gains and losses in our sample of property listings, computed as the percent difference between the estimated hedonic value of the property at the time of listing, and the initial purchase price. Overall, this “potential gain” distribution for the full set of listings in the data suggests that approximately 25% of sellers face nominal losses if they were to sell at hedonic value. The share of owners with potential paper losses varies between 12.8% and 40.8% (Panel B) across regions of the U.K. which suggests that the behavioral motive is operative for a sizable fraction of the population.

For properties that are financed with a mortgage, the “potential home equity” position of the seller is calculated as the difference between the outstanding mortgage amount and the hedonic value at the time of listing. In the U.K. mortgage market, interest costs start to increase for loan-to-value ratios above 80-85% (Liu, 2022). This implies that a seller with home equity below 20% will find it more costly to take a loan to finance purchasing a property of similar value to the one they are selling, and may have to consider downsizing. That said, a sizable fraction of all mortgages are issued with *initial* LTVs above 80%, which suggests that for most

¹⁰Some of the variables used in the regression are not available in our data for all properties. Most importantly, we frequently miss information on the number of bathrooms. Therefore, we impute the number of bathrooms by regressing it as a continuous variable on floor area with fixed effects on number of bedrooms, electoral ward, and property type, and dummy variables on whether the property is a retirement home, auctioned, a development, or affordable housing property. The hedonic model that we use in our subsequent analysis uses imputed number of bathrooms to handle the missing data. The model fit across different price levels is shown in Figure A.3.

mortgage borrowers, downsizing is only expected to be a material possibility when they borrow at far greater LTV ratios.

In the data, we estimate that a tiny fraction (3%) of all outstanding mortgage contracts across all time periods are underwater, and only a relatively small share (16%) have a home equity level below 20%. Despite these apparently low magnitudes, there is a theoretically distinct role for down-payment constraints for both seller strategy and final transaction outcomes (Stein, 1995). Therefore, we explicitly incorporate such downsizing/credit financing frictions in our theoretical and empirical analyses.

3 Motivating facts

3.1 Housing market transaction volume

It is well-known that housing market liquidity is lower when house prices fall and higher when prices rise (Genesove and Mayer, 2001; Ngai and Tenreyro, 2014). In Table 1, we estimate a standard geometric repeat sales model (Bailey, Muth and Nourse, 1963) on repeated transactions of properties since 1995 using the universe of housing transactions reported by the Land Registry. In each month, we assign each property in the data to either the gain or loss domain depending on whether the repeat sales price index has increased or decreased since the last transaction on the property. This allows us to build a monthly aggregate loss share measure on the *whole stock of properties*. The estimation is implemented from 2005 onwards to allow for a 10-year build-up of transactions in the data from which the loss share is calculated. We replicate this analysis in the U.S. housing market using price and volume indices from Zillow, starting in 2008, and collecting a representative sample of purchase dates collating all the versions of the Survey of Consumer Finances since 2000. For each month, and for each purchase date preceding that month in any SCF wave, we assign the purchase to loss or gain domain, depending on whether the state-level Zillow price index has decreased or increased since purchase. Consistently in U.S. and U.K. data, we find that the loss share measure explains a large part (up to 70%) of the observed price–volume elasticity. Building on these results, our framework provides a conceptually novel approach to interpret the dynamics of transaction volumes in the housing market, replacing a mechanism that relies on price changes to one that emphasizes the distribution of nominal reference points in the population.

Table 1
Drivers of housing market transaction volume: International comparison

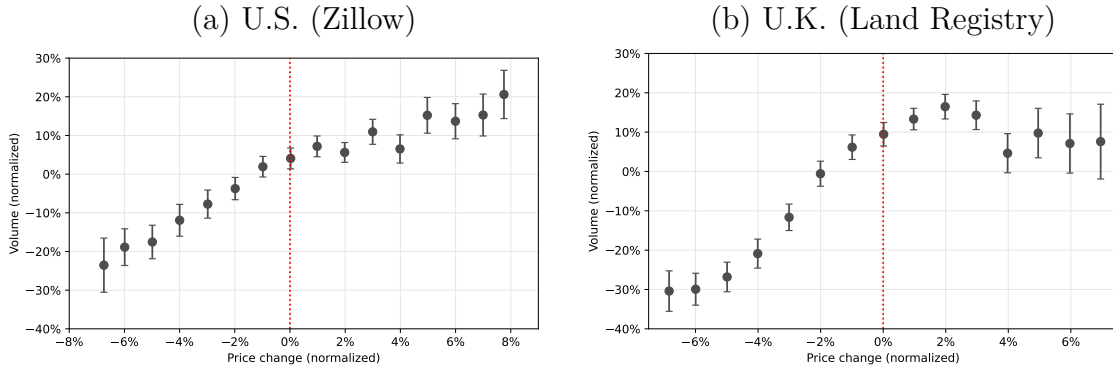
This table reports the results for a monthly regression of log transaction volume on contemporaneous price growth rates with controls for aggregate loss shares. The analysis uses repeat sales price indices using only Land Registry transactions data for U.K. and Zillow price and volume data and purchase dates from the Survey of Consumer Finances (SCF) for the U.S. The loss share measure is calculated monthly for the whole nationwide stock of properties, which have at least one transaction in the data preceding the month at which the measure is calculated. Properties for which the aggregate sales index has decreased since the time of purchase are considered having paper losses. For the case of the U.K., the sample starts in 2005, and for the U.S. it starts in 2008. *, **, *** indicate statistical significance at the 10%, 5% and 1% level, based on standard errors clustered at the level of ITL2 regions in the U.K. and states in the U.S.A.

	<i>Dependent variable: Transaction volume (log)</i>							
	U.S. (Zillow and SCF)				U.K. (Land Registry)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Price growth	2.42*** (0.220)		0.751*** (0.135)		2.39*** (0.107)		0.508* (0.190)	
Loss share		-0.951*** (0.060)	-0.829*** (0.063)	-1.18*** (0.094)		-1.70*** (0.064)	-1.54*** (0.108)	-1.54*** (0.077)
Share of gains in [0%, 20%]				0.182* (0.078)				0.565*** (0.052)
ITL2 / State FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,927	5,927	5,927	5,927	7,595	7,595	7,595	7,595
R ²	0.910	0.929	0.930	0.929	0.167	0.325	0.330	0.347

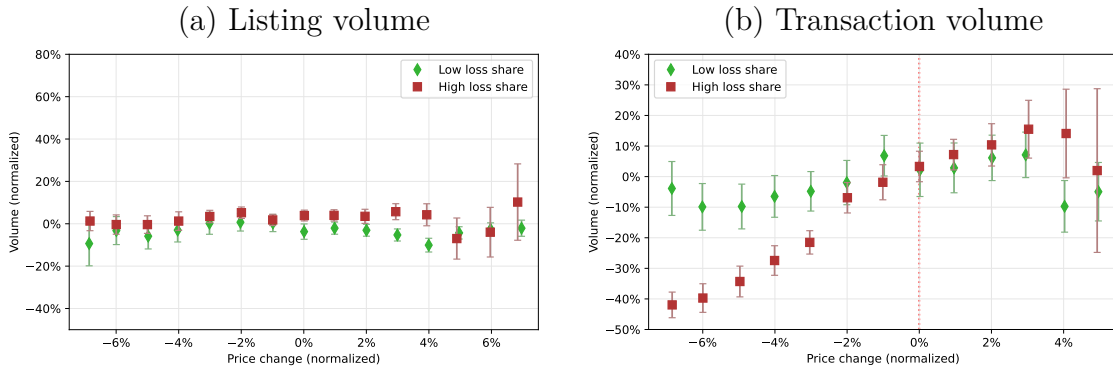
Figure 1
Comovement of prices and volumes

Panel A reports the correlation between year-over-year changes in housing prices and the volume of realized transactions. We calculate prices and volumes at the level of ITL2 regions (U.K.) and the level of states (U.S.), respectively, measured monthly for the period between January 2010 and December 2022. We express the volume series as a relative deviation to the location-specific mean and the price growth as a relative percent difference to the location-specific mean. On the horizontal axis, we report price growth rate bins, constructed by rounding to the second decimal, for a domain restricted to $[-8\%, 8\%]$ for comparability. Online appendix Figure A.4 shows the corresponding frequency distribution. Panel B reports the correlation between year-over-year changes in housing prices and the volume of listings (i) and realized transactions (ii). We distinguish between periods and regions for which the share of sellers that face the possibility of a loss is below/above the median in the sample (“Low/High loss share”), respectively. Error bars indicate 95% confidence intervals for bin-specific means.

Panel A
Aggregate effects



Panel B
The role of the loss share



We further illustrate this relationship between prices and transaction volumes at the level of ITL2 regions of the U.K. and states of the U.S. at the monthly frequency between January 2010 and December 2022. To do so, we construct a price growth series using the location $\times$ month fixed effects from the hedonic pricing model in equation (1), and compute the corresponding number of transactions from HM Land Registry for each month and location. For the U.S., we use the median transaction price and the reported number of residential property sales in each state and month. For each location-month observation, we express the volume series as a relative deviation to the location-specific mean and price growth as a relative per cent difference to the location-specific mean.

Using these data, Panel A of Figure 1 shows a robust positive price-volume comovement in both markets. The estimated magnitude of the comovement is economically very significant, with an 8% location-month-specific price increase associated with a 20% increase in corresponding location-month-specific transaction volume in the U.S. In the U.K., an 8% price decrease is associated with a corresponding 30% drop. To more precisely quantify the size of the comovement, we estimate the following panel regression:

$$\ln V_{i,t} = \gamma_i + \beta_P \Delta \ln P_{i,t} + \varepsilon_{i,t}, \quad (2)$$

where $V_{i,t}$ is the normalized number of transactions, $\Delta \ln P_{i,t}$ is the year-over-year growth rate of log prices in location i and month t , and γ_i is a location-specific fixed effect.

Does the extent of nominal anchoring affect these and other aggregate relationships in the housing market? In Figure 1 we calculate the share of sellers with nominal losses in each month in each ITL2 region of the U.K. We then compare the price-volume comovement in region-months where the share of nominal losses lie above the in-sample mean (“High” in the plot’s legend), with those region-months where the share of nominal losses lies below the in-sample mean (“Low” in the plot’s legend). We lag the classification of region-months by one month relative to the time at which we measure price growth and volumes to avoid any mechanical correlation. The figure shows that the positive relationship between prices and volumes is far more pronounced when the share of homeowners facing nominal losses is high, and much more muted when a relatively lower fraction of the population faces nominal losses relative to historical reference points. In terms of magnitudes, we find that the

unconditional increase in volumes associated with a 1% increase in prices is equal to 1.05%, and this number rises to 3.16% when the “paper loss” share is high.

3.2 Prices and volumes across regions

For a more granular depiction of this result, in Panel A of Figure 2 we estimate the price–volume comovement in equation (2) separately for each ITL2 region of the UK:

$$\ln V_{i,t} = \gamma_i + \beta_{i,P} \Delta \ln P_{i,t} + \varepsilon_{i,t}, \quad (3)$$

and plot the estimated location-specific coefficients $\beta_{i,P}$ against the share of properties that face nominal losses in each corresponding region, the loss share calculated from the full sample of listings in 2010 – 2022. The plot shows a positive association, with a univariate $R^2 = 0.30$. This is further suggestive evidence for the potential role of behavioral lock-in, if we interpret this as the elasticity of volumes with respect to price declines being more pronounced in regions where a large number of prospective house sellers face nominal losses.

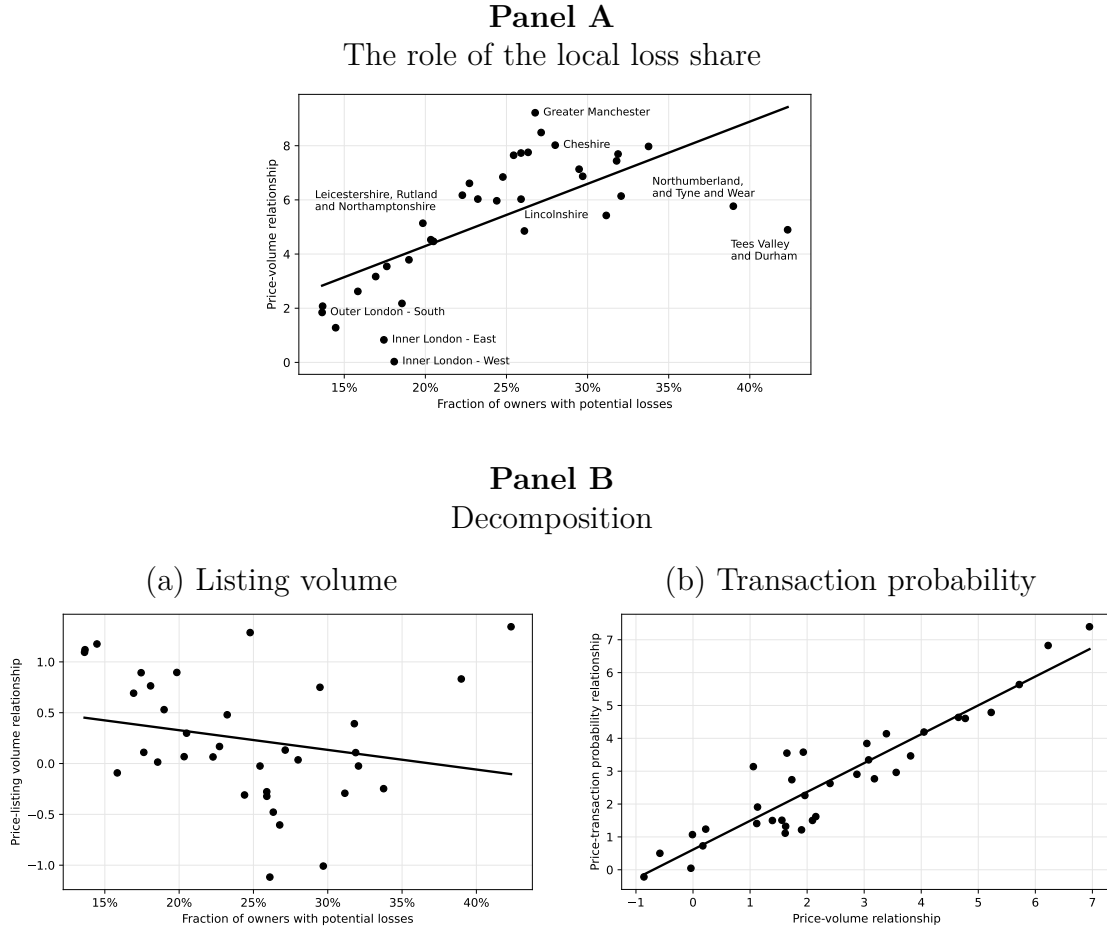
In column 1 of Table A.1, we report the relationship between the change in the volume of houses listed to year-on-year price growth rates in the U.K. (i.e., capturing the aggregate response of the extensive margin decision of households to list properties to price movements). Column 2 of the table shows that this comovement is dwarfed by the relationship between the change in transaction volumes *conditional on listing* and price growth rates (i.e., capturing the intensive margin change in the rate at which already listed properties are taken off the market in response to price changes). This result shows that the majority of the price-volume relationship comes from faster turnover of already listed properties when there are price growth shocks, whereas the response of listing volumes is more muted. Put differently, since the total price-volume comovement is fully decomposed into these two extensive and intensive margin components, our new finding is that the intensive margin (faster turnover of already listed properties) is the dominant channel through which the price-volume relationship arises, while the response of listing volumes to price changes is relatively more muted.

To explore this finding further, Panel B of Figure 2 considers the role of extensive and intensive margins in explaining the price–volume relationship within each ITL2 region. The figure shows that not only across time, but also across regions, the

response to price changes of the conversion rate of listings to sales is the main driving factor for the cross-regional variation in comovement between total housing volumes and price changes.

Figure 2
Cross-regional variation in the price-volume relationship

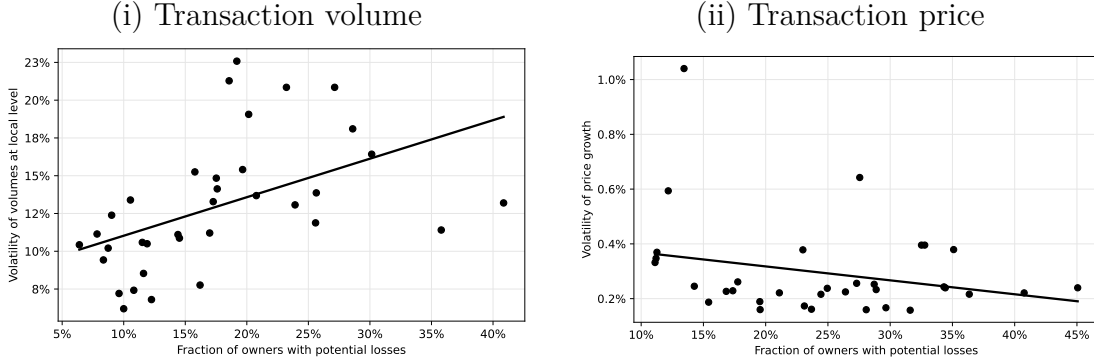
Panel A plots regression coefficients from regressions of the realized transaction volume on the year-on-year price change in each ITL2 region, for the period between January 2010 and December 2022. The horizontal axis indicates the share of owners that list a property for sale, for which the estimated hedonic value of the property is below the initial purchase price. Panel B calculates corresponding region-specific regression coefficients where the dependent variable is, respectively, (i) the listing volume, and (ii) the ratio between the transaction volume and the listing volume, i.e., the transaction probability.



Finally, Figure 3 reports the time-series volatility (i.e. standard deviation) of volumes and price growth rates across ITL2 regions, plotted against the regional share of potential nominal losses. In regions where the potential loss share is high,

Figure 3
Cross-regional variation in volatilities

The figure reports region-specific volatilities of transaction volumes and price growth rates, measured as the standard deviation of realized sales volume in each ITL2 region over the period between January 2010 and December 2022. The horizontal axis indicates the share of owners that list a property for sale, for which the estimated hedonic value of the property is below the initial purchase price.



the time series volatility of volumes is substantially higher, and the volatility of prices is slightly lower. This finding aligns with the broad intuition about the role of a behavioral nominal rigidity—in areas where many agents are reluctant to realize nominal losses when housing values decline, volumes collapse and absorb part of the price volatility, and vice versa when housing values rise and gains occur. Volumes thus react more strongly than prices to shocks affecting the housing market.

3.3 Regional heterogeneity: Bunching of transaction prices

[Andersen et al. \(2022\)](#) use micro administrative data from Denmark to identify the nominal purchase price of a property as a quantitatively relevant reference point for sellers of residential properties. They test for reference-dependent preferences using a joint set of *ex ante* and *ex post* empirical moments that capture the behavior of sellers throughout the transaction process. They rationalize the empirical patterns using a model in which preferences exhibit reference dependence and loss aversion, with a change in slope (a “kink”) at the reference point.

In Figure 4, we verify that such patterns are also evident in U.K. micro housing transactions data. The left hand panel of the figure shows the distribution of nominal gains, computed as the difference between the observed re-sale transaction price and the original purchase value. To extract a measure of “excess mass” from the

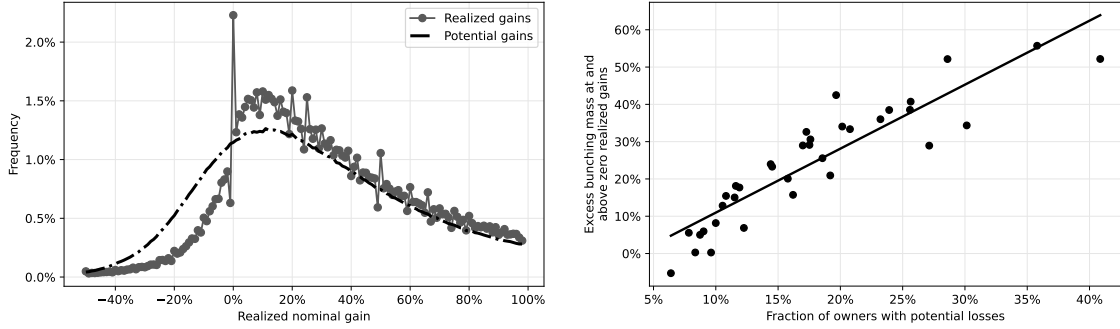
distribution of realized transaction prices, we subtract from this distribution the counterfactual frequency of nominal gains that would occur if properties were sold exactly at their estimated hedonic values. Consistent with prior evidence, we find missing mass immediately below the reference point and significant bunching at exactly zero nominal gains. A build-up of additional mass is observed as diffuse bunching for nominal gains in the low positive domain, which is consistent with reference dependence in the presence of optimization frictions ([Anagol et al., 2022](#)).

The right hand panel plots this measure of excess mass for each U.K. ITL2 region (on the y-axis) against the share of sellers who face nominal losses in each of these regions. There is a strong correlation between the two measures. In other words, the extent of excess bunching mass to the right of the reference point—generated by active reference-dependent seller behavior—is tightly linked to the size of the overall group of sellers that face nominal losses. This reinforces our interpretation that reference dependence is the key driver of the results we have documented. The share of sellers facing nominal losses measures the extent to which the behavioural force is operative in each ITL2 region.

Having linked the loss share empirically to a number of aggregate housing market outcomes, we next build a dynamic housing market model to provide a more rigorous theoretical foundation for these observations, and aid the evaluation of housing market policy.

Figure 4
Excess bunching mass

The left hand plot reports the distribution of differences between the realized price in a given sale transaction and the price for which the property has been initially purchased (the nominal realized gain), and compares it with the counterfactual distribution of nominal potential gains among the same set of properties. We use binned percentage point intervals that are closed to the left and open to the right (i.e., the 0% bin contains gains in the interval between 0% and strictly below 1%). The right hand plot reports the total relative excess mass for nominal realized gains between 0% and 20% for each ITL2 region separately, and plots this quantity against the average “loss share” in that region, i.e., the share of owners for which the estimated hedonic property value is below the initial purchase price.



4 The Model

Motivated by the facts discussed in Section 3, we develop a dynamic heterogeneous agent search and matching model of the housing market where homeowners are reference dependent and loss-averse around a pre-determined anchor, and face realistic financial constraints. Seller decisions are conditional on heterogeneous realizations of shocks to mobility, heterogeneous levels of reference prices, and heterogeneous outstanding mortgage amounts. Buyer decisions are conditional on heterogeneous realizations of housing matching quality shocks. The participation and pricing decisions of homeowners in the search and matching process determine aggregate housing market outcomes including prices and transactions volumes.

4.1 Environment

Consider a discrete-time formulation of an economy with a housing stock N_H potentially available for sale, and a mass N_B of interested buyers. Current homeowners choose whether to list their houses for sale, and buyers randomly search over the outstanding stock of available listings. We denote the endogenous number of sellers in a generic period t by N_{St} and the resulting market tightness by $q_t = N_B/N_{St}$.¹¹ The probability that a seller meets a buyer is modelled using a constant returns to scale matching function $\chi(q_t)$, which we specify as $\chi(q_t) \equiv \bar{\chi} \cdot q_t^v$, for some constants $\bar{\chi} > 0$ and $v \in [0, 1]$. The probability that a buyer meets a seller is then equal to $\chi(q_t)N_{St}/N_B = \chi(q_t)/q_t$. Upon a successful transaction, the buyer becomes a homeowner and the seller begins their search for a new home as a buyer, which implies that the overall number of buyers N_B remains constant through time.¹²

4.2 Homeowners

Owning a home generates flow utility u_t . At the beginning of period t , each homeowner i draws a mobility shock $\theta_{it} \sim F_\theta(\cdot)$, which represents a newly available outside opportunity with a payoff that only gets realized if the house is successfully sold. Given the draw, the homeowner either lists the property for sale and pays a one-time

¹¹The market becomes “tighter” when the number of buyers searching actively increases relative to the number of sellers, and is less tight when the opposite is true.

¹²Thus a positive demand shock leads households to “move up the housing ladder”, and individual decisions spill over to the broader housing market (Aiello et al., 2022).

cost $\phi \geq 0$ for doing so, or they ignore the opportunity, and take another independently distributed draw during the next period. If the homeowner chooses to list, they choose a log list price l_{it} .

With probability $\alpha_t(l_{it})$, the sale transaction completes successfully at price $p_{it} \equiv p(l_{it})$ and the seller receives a one-time utility payoff $U(p_{it}, r_i, m_{it}) + \theta_{it}$, where r_i denotes the log *reference price* for which they originally purchased their property, and m_{it} their log mortgage balance; and they start searching for a new property, from the position of a new buyer. With probability $1 - \alpha_t(l_{it})$, the homeowner is unable to sell their property, and they draw another independent and identically distributed realization of the mobility shock in the next period.

We assume that the function $p(\cdot)$ transforming listing prices to transaction prices is deterministic and common knowledge. Therefore, the listing price l_{it} is isomorphic to a take-it-or-leave-it offer at price $p(l_{it})$. In our empirical analysis, we estimate $p(\cdot)$ directly from the data, but for now note that the shape of this function affects sellers' price setting incentives.

We denote the homeowner's value function by $V_t^h(r_i, m_{it})$ and define it through the following Bellman equation:

$$V_t^h(r_i, m_{it}) = u_t - \tau^h + \mathbb{E}_\theta \max \left\{ \underbrace{\max_{l_{it}} \alpha_t(l_{it}) [U(\cdot) + \theta_{it} + \beta \mathbb{E}_t[V_{t+1}^b]]}_{\text{Listing success}} \right. \\ \left. + \underbrace{(1 - \alpha_t(l_{it})) \beta \mathbb{E}_t[V_{t+1}^h(\cdot)] - \phi}_{\text{Listing failure}}, \underbrace{\mathbb{E}_t[\beta V_{t+1}^h(\cdot)]}_{\text{No listing}} \right\}, \quad (4)$$

where $\tau^h = \ln \left(\frac{1}{1 - \text{property tax}} \right)$.

4.2.1 Seller utility

We assume the seller utility function includes reference dependence and loss aversion, and set:

$$U(p_{it}, r_i, m_i) = p_{it} + \underbrace{W(p_{it}, r_i)}_{\text{Reference dependence}} - \underbrace{\mu(\gamma - (p_{it} - m_i))_+^2}_{\text{Downsizing penalty}}, \quad (5)$$

where:

$$W(p_{it}, r_i) = \begin{cases} \eta(p_{it} - r_i), & \text{if } p_{it} \geq r_i, \\ \lambda \eta(p_{it} - r_i), & \text{if } p_{it} < r_i. \end{cases} \quad (6)$$

The function $W(\cdot)$ describes reference dependent ($\eta > 0$) and loss averse ($\lambda >$

1) preferences. We also include a convex downsizing penalty as mentioned above, motivated by [Stein \(1995\)](#) and [Andersen et al. \(2022\)](#). If the seller's home equity is lower than the exogenously set threshold γ , they expect to face additional costs to finance another house of similar size or quality after selling their home. Therefore, they dislike the prospect of selling at a price that reduces home equity beneath the threshold, because that would require either painful down-sizing if they cannot acquire financing, or costly financing to bridge the gap.

4.2.2 Homeowners' extensive margin decision

Conditional on listing, the maximization problem for the list price l produces an optimal list price $l_t^*(r, m, \theta)$ and corresponding transaction price $p_t^*(r, m, \theta)$ for each pair of state variables r and m , and for each draw of θ . The extensive margin decision is governed by a threshold rule, where homeowners list their property for sale if and only if:

$$\alpha_t(l_t^*(r_i, m_{it}, \theta_{it})) \left[U(p_t^*(r_i, m_{it}, \theta_{it}), r_i, m_{it}) + \theta_{it} + \beta \mathbb{E}_t[V_{t+1}^b] - \beta \mathbb{E}_t[V_{t+1}^h(r_i, m_{it})] \right] \geq \phi.$$

This yields a threshold rule for listing:

$$\theta_{it} \geq \frac{\phi}{\alpha_t(l_{it}^*)} - [U(p_{it}^*, r_i, m_{it}) + \beta \mathbb{E}_t[V_{t+1}^b] - \beta \mathbb{E}_t[V_{t+1}^h(r_i, m_{it})]] \equiv \theta_{it}^*(r_i, m_{it}), \quad (7)$$

where we have denoted $l_{it}^* \equiv l_t^*(r_i, m_{it}, \theta)$, $p_{it}^* \equiv p_t^*(r_i, m_{it}, \theta_{it}^*)$. Notice that the equality in (7) is a consistency condition for θ_{it}^* . Together with $p_{it} = p(l_{it})$ and the optimal price setting condition from the seller's optimization problem, these constitute three equations on three unknowns l_{it}^* , p_{it}^* and θ_{it}^* for each (r_i, m_{it}) pair.

4.3 Mortgage amortization

After a transaction between a seller and a buyer at price R_i , the buyer draws a random mortgage balance $M_{i,t+1} \sim F_M([0, R_i])$ and enters the economy as a homeowner with state $= (\ln R_i, \ln M_{i,t+1}) = (r_i, m_{i,t+1})$. Afterwards, we assume the mortgage loan-to-value ratio (relative to the original purchase price), denoted $\tilde{m}_{i,t+1}$, reduces deterministically: $\tilde{m}_{i,t+2} = \tilde{m}_{i,t+1} - \delta_m$, for a constant δ_m . This implies that the mortgage amortization rate is independent of the original purchase price or issuance LTV.

4.4 Buyers

Each period, buyers search for houses randomly. Upon meeting with a seller i , buyer j draws an idiosyncratic taste shock $\varepsilon_{jt} \sim F_\varepsilon(\cdot)$, which leads them to accept the sales offer and to purchase the property, if and only if:

$$\beta \mathbb{E}_t[V_{t+1}^h(p_{it}, m_{j,t+1})] + \varepsilon_{jt} \geq p_{it} + \tau^b + \beta \mathbb{E}_t[V_{t+1}^b],$$

where V_t^b is the buyer's value function at time t , p_{it} is the log transaction price of the sale (associated with the log listing price l_{it} set by the seller i), τ^b is the log of (one plus) the stamp duty tax rate on the purchase, and $m_{j,t+1}$ is the log mortgage amount taken to purchase the property. The transaction price becomes the buyer's reference price as they become a new homeowner upon sale completion.

The buyer's optimal choice is thus also governed by a threshold rule:

$$\varepsilon_{jt} \geq p_{it} + \tau^b + \beta \mathbb{E}_t[V_{t+1}^b] - \beta \mathbb{E}_t[V_{t+1}^h(p_{it}, m_{j,t+1})] \equiv \varepsilon_t^*(p_{it}). \quad (8)$$

Note that both taste shocks and seller and buyer values are measured in units of log price. Therefore, the model implicitly assumes log utility on transaction prices for buyers and sellers.

The buyer's Bellman equation can then be expressed as:

$$V_t^b = \underbrace{\beta \mathbb{E}_t[V_{t+1}^b]}_{\text{Value of remaining as buyer}} + \frac{\chi(q_t)}{q_t} \int_l \int_{\varepsilon_t^*(p(l))}^{\infty} \underbrace{[\varepsilon - \varepsilon_t^*(p(l))]}_{\text{Option value for high taste}} d\Phi_\varepsilon(\varepsilon) d\Omega_t(l). \quad (9)$$

where $d\Omega_t(\cdot)$ is the endogenous density of available list prices.

4.5 Aggregation and model dynamics

We denote the (time-varying) density of homeowners by $f_t(\cdot, \cdot)$ over reference prices r and mortgages m_t . At each (r, m_t) pair, each existing homeowner draws a mobility shock θ_t , they choose whether to list the property for sale, and choose the corresponding asking price $l_t^*(r, m_t, \theta_t)$. These optimal choices induce an endogenous listing price distribution, which we denote by $d\Omega_t(l) = \omega_t(l) dl$.

At each reference price r and mortgage balance m_t , the share of existing homeowners who decide to list their property for sale is $1 - F_\theta(\theta_t^*(r, m_t))$; with probability

$\alpha_t(l_t^*(r, m_t, \theta))$, sales go through. This is endogenously determined by the number of meetings and the probability of buyers accepting their offers, conditional on a meeting:

$$\alpha_t(l) = \underbrace{\chi(q_t)}_{\text{Probability that seller meets a buyer}} \times \underbrace{[1 - F_\varepsilon(\varepsilon_t^*(p(l)))]}_{\text{Probability of acceptance conditional on a meeting}}. \quad (10)$$

On the other hand, after a successful transaction, the new homeowner draws a random log mortgage balance and enters the economy with reference price r and mortgage balance $m_{t+1} \sim F_m(\cdot)$. The mass of new homeowners with reference price r and mortgage balance m_{t+1} is¹³

$$\underbrace{N_{St}\omega_t(r)}_{\text{Number of sellers who list at price } r} \times \underbrace{\alpha_t(r)}_{\text{Probability of sale conditional on listing}} \times \underbrace{f_m(m_{t+1})}_{\text{Density of new homeowners with mortgage } m_{t+1}},$$

where we abuse the notation by denoting by $\omega(r), \alpha(r)$ the listing density and probability of transaction, respectively, for the listing price l , which leads to the transaction at price $r(l)$, should the buyer accept the offer.

Taken together, it follows that the law of motion for the homeowner distribution is given by:

$$\begin{aligned} f_{t+1}(r, m_{t+1}) &= \text{Existing homeowners not selling} + \text{New homeowners} \\ &= \underbrace{f_t(r, m_t)}_{\text{Existing homeowners}} \times \underbrace{\left[1 - \int_{\theta_t^*(r, m_t)}^{\infty} \alpha_t(l_t^*(r, m_t, \theta)) d\Phi_\theta(\theta)\right]}_{\text{Non-sellers and failed sellers}} \\ &\quad + \underbrace{N_{St}\omega_t(r)\alpha(r)f_m(m_{t+1})}_{\text{New homeowners}}, \end{aligned} \quad (11)$$

where $m_{t+1} = \ln(M_{t+1})$ and $M_{t+1} = M_t - R \cdot \delta_m$, i.e., the mortgage is reduced by a constant fraction of the purchase price R .

¹³To simplify the clutter, we abuse the notation here by replacing the listing price l with the associated sales price $p(l)$, which becomes the buyer's reference price $r = p(l)$ upon the transaction. I.e., each listing price is associated with a deterministic transaction price $p(l)$ upon a transaction and the demand function and listing distribution can also be considered w.r.t. these transaction prices. For details, we refer to Appendix A.9.

Finally, we close the model with

$$N_{St} = \int_{r,m} \int_{\theta^*(r,m)}^{\infty} f_{\theta}(\theta) d\theta dF_t(r, m), \quad (12)$$

which constitutes a fixed point problem, since the optimal threshold rule for listing depends on aggregate equilibrium outcomes.

4.6 Equilibrium

We specify the distribution of the seller and buyer shocks as normal distributions $F_{\theta}(\cdot) = \Phi(\cdot; \mu_{\theta}, \sigma_{\theta})$ and $F_{\varepsilon}(\cdot) = \Phi(\cdot; \mu_{\varepsilon}, \sigma_{\varepsilon})$, and define the equilibrium concept as follows.

Definition. An equilibrium consists of:

- buyer and seller value functions $V_t^b, V_t^h(r, m)$ defined through the Bellman equations (9) and (4), respectively;
- buyer policy function $\varepsilon_t^*(p)$ satisfying (8);
- seller price setting policy $l_t^*(r, m, \theta)$, which solves the seller's price setting problem, and extensive margin policy function $\theta_t^*(r, m)$, which solves (7);
- aggregate transaction probability $\alpha_t(l)$ satisfying (10);
- distribution of reference prices and mortgage balances $F_t(r, m)$, whose law of motion satisfies (11);
- distribution of listing prices $\Omega_t(l)$, which arises endogenously from the sellers' extensive and intensive margin decisions; and
- market tightness q_t satisfying (12).

In our analyses, we concentrate on studying the stationary equilibrium with $F(\cdot, \cdot) \equiv F_t(\cdot, \cdot) = F_{t+1}(\cdot, \cdot)$, $V^b \equiv V_t^b, V^h(\cdot, \cdot) \equiv V_t^h(\cdot, \cdot)$ and so forth, where the equilibrium quantities do not explicitly depend on time.

4.7 Government tax revenue

While we do not explicitly model the government, we note that the presence of both transactions taxes and ongoing property taxes in the model facilitates our later work on understanding how fiscal policy related to housing is affected by the presence of behavioral frictions.

In our setup, we assume that the government collects taxes from both the stamp duty transactions tax and the council tax (ongoing property taxes in the U.K.). Consistent with the institutional setting, we model the council tax as a property tax proportional to the average value of the house. Government total tax revenue is therefore given by:

$$T = \text{Stamp duty revenue} + \text{Property tax revenue}$$

$$= \underbrace{[\exp(\tau^b) - 1]}_{\text{Stamp duty rate}} \times \underbrace{N_{St} \int \alpha_t(p) \exp(p) d\Omega_t(p)}_{\text{Total value of sales}} + \underbrace{[1 - \exp(-\tau^h)]y}_{\text{Property tax rate}} \times \underbrace{\frac{\int \alpha_t(p) \exp(p) d\Omega_t(p)}{\int \alpha_t(p) d\Omega_t(p)}}_{\text{Average transaction price}}.$$

Note that τ^h is defined as a flow utility cost out of consumption. To translate this to a property tax paid as a percentage of the house value, we need to scale it by the rental yield y , which we assume is exogenous.

4.8 Model features

4.8.1 Endogenous disagreement

The value of a property for a homeowner is $V_t^h(r, m)$. This consists of the utility flow u from owning the property and the option value associated with good outside opportunities θ .

When a buyer considers buying a property, they compare the price p_{it} to the expected property value $\beta \mathbb{E}_t[V_{t+1}^h(p_{it}, m_{t+1})]$ and their idiosyncratic taste ε_{jt} for the property. The seller in turn considers the price p_{it} they receive, their value $\beta \mathbb{E}_t[V_{t+1}^h(r_i, m_{it})]$ for keeping the property, and their idiosyncratic mobility shock θ_{it} .

Apart from idiosyncratic buyer taste ε_{jt} and seller outside option θ_{it} , the trade-offs the buyer and seller face are between the price p_{it} and the buyer and seller property valuations $\beta \mathbb{E}_t[V_{t+1}^h(p_{it}, m_{t+1})]$ and $\beta \mathbb{E}_t[V_{t+1}^h(r_i, m_{it})]$, respectively. This difference in the property valuations between the buyer and the seller generates disagreement in terms of what constitutes property “fair value”. In other words, there is no mutually

agreed fair price for the property. Despite this gap between WTA and WTP, the buyer’s individual taste and the seller’s outside option can facilitate trade; i.e., the gap between the property valuations is closed by the individual circumstances of the agents. Put differently, the seller may be willing to accept a lower price because of a good outside option (driven by θ), or the buyer may accept a higher price due to their idiosyncratic taste for the particular property (driven by ε) even if they consider it to be overpriced.

When conducting empirical tests of the model predictions in section 5, we think of the listing premium in the model as the difference of the listing price and a measure of the average transaction price without taking a stance on the notion of fair value (given that buyers and sellers disagree on this notion in the model). More specifically, this corresponds to interpreting the hedonic valuation that we compute in the data (which we subtract from the listing price to compute the empirical listing premium) as an average price for the particular property with given hedonic characteristics.¹⁴

4.8.2 Rational behavioralism

The difference of opinion between buyers and sellers in the model stems from two sources. On the one hand, sellers’ valuations depend on their reference prices, which are unique to each seller. On the other hand, buyers understand that whatever price p they pay for the property will become their reference point. This will affect their future sales behavior, should they ever want or need to sell the property. In other words, when paying a high price, the buyer understands that it will be challenging to sell the property at that price later, and they incorporate this prospect in their decisions *ex-ante*. Thus, buyers anticipate their future reference-dependent behavior and discount future outside options (i.e., mobility shocks θ) accordingly.

When facing a positive demand shock that drives prices upwards, sellers have better prospects on the market but they also anticipate the need to pay higher prices when they become buyers in the future. Therefore, the quantitative effect of a positive demand shock depends on the relative magnitude of each of these forces. Given that purchasing a property is always the buyer’s free choice, and a seller is never forced to buy another property in the future, we expect sellers to respond

¹⁴In the model all properties are considered identical, whereas in the data the properties have heterogeneous hedonic characteristics. In our empirical tests, we construct price indices using the time fixed-effects from the hedonic model. These can be considered as average valuations cleaned of heterogeneity in property characteristics.

positively to positive demand shocks. However, the quantitative effect is reduced by their expectations of higher purchase prices in the future. In case of a negative shock, the response is also asymmetric since, due to loss aversion, sellers dislike the nominal losses from selling the property more than they like the prospect of buying cheap in the future. In conclusion, all agents in the model have rational expectations, and the behavioral component is only due to the preference structure that we assume, a situation that we label as *rational behavioralism*.

4.8.3 Bargaining

In terms of realized sales prices, the model assumes that on the intensive margin sellers have all the bargaining power, through *de facto* take-it-or-leave-it offers, since the function $p(\cdot)$ transforming listing prices to transaction prices is common knowledge.¹⁵ This means that for any price negotiation process that takes place, both the sellers and the buyers can perfectly anticipate the end result of such bargaining, and that result is described by the function $p(\cdot)$. However, the model does include bargaining through the extensive margin decisions of sellers and buyers. Buyers exercise their market power in terms of their choice of whether or not to accept a seller's offer, as opposed to the intensive margin of bargaining over the final price. Similarly, sellers' listing prices depend on their outside options: the optimal list price is a function of θ_{it} as $l_{it}^* = l_t^*(r_i, m_{it}, \theta_{it})$. Sellers with good outside options set lower prices, since the opportunity cost of being unable to sell is higher than for those sellers with a relatively poorer outside option.

4.8.4 Location as an asset

Finally, we note that the value $V_t^h(r, m)$ of a house from the perspective of the homeowner includes the expectation of receiving high outside options in the future. From the buyers' perspective, these options are available only if a property is purchased. Therefore, we can interpret the choice of a property as the choice of a city/location with good outside options, similar to [Bilal and Rossi-Hansberg \(2021\)](#). Conversely, sellers understand that selling their property could lead to utility gains through improved potential future matching outcomes, captured by the shock ε .

¹⁵We define *common knowledge* formally: a fact is said to be common knowledge among a set of agents when all agents know the fact, they all know that they know the fact, they all know that they all know that they know the fact, and so on *ad infinitum*.

5 Model estimation

We use our highly detailed microdata to estimate a number of key model parameters. This approach builds on a large and growing micro literature that documents the effects of reference dependence and loss aversion in the housing market. In our estimation, we target three facts that are well-known in the existing literature ([Andersen et al., 2022](#); [Guren, 2018](#); [Genesove and Mayer, 2001](#)) and documented in Figure 5: (i) listing premium hockey stick: sellers who face nominal losses substantially mark up their listing prices over the hedonic value of the property, consistent with preferences that are reference-dependent and loss-averse; (ii) home equity hockey stick: financially constrained sellers similarly mark up their listing prices in an attempt to avoid costly downsizing; and (iii) concave demand: the probability of a transaction going through decreases in the listing price, but is bounded above for low or negative listing premia.

We use these micro patterns in the U.K. data as targets in our estimation exercise, which we describe in detail below. We then evaluate the ability of the resulting calibrated model to explain a set of untargeted macro moments.

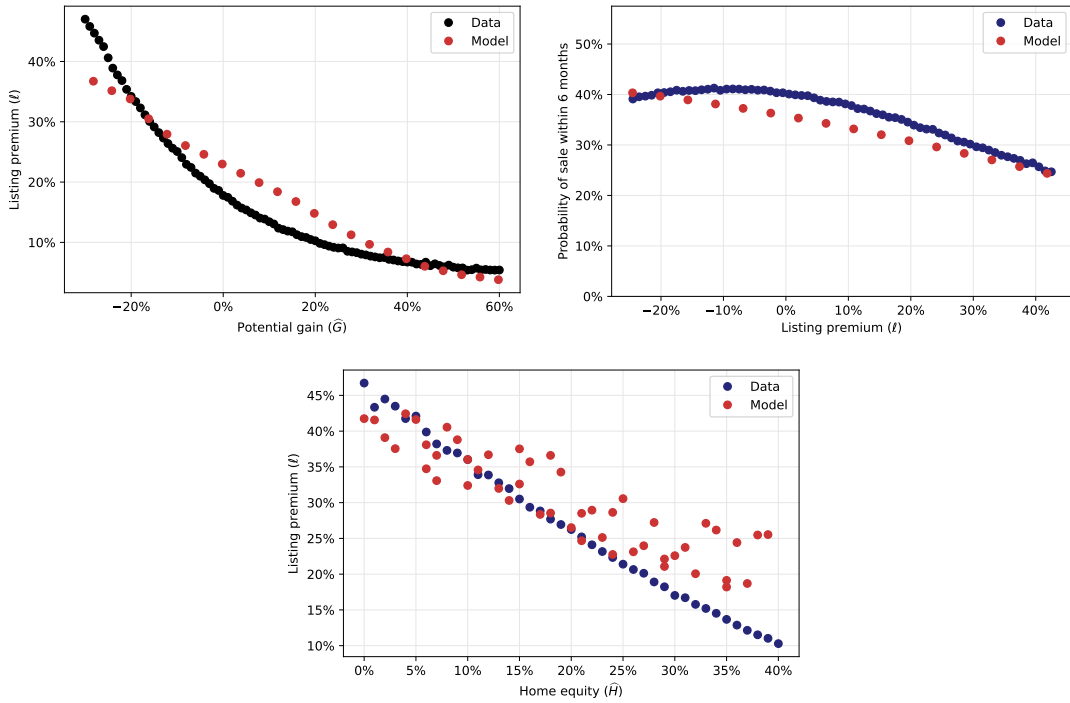
5.1 Externally calibrated parameters

We solve the model numerically, using value function iteration to solve for a deterministic steady state. We define a time period in the model to correspond to 6 months in the data, and set the time discount factor equal to $\beta = 0.99$ (i.e., we assume an annualized real discount rate of 2%). For a convenient interpretation of aggregate quantities, we normalize the housing stock to $N_H = 1$ and the average log price level to 1. This means that all volumes in the model are relative to the housing stock, and all prices are relative to the price of a representative property. The top part of Table 2 lists the parameter values that we calibrate independently from the structural estimation. The bottom part of the table shows parameters that reflect the institutional and policy environment in the U.K.

For the calibration of the aggregate matching function, we use observed listings and search behavior during a sub-period in our sample, i.e., between 2018 and 2022. We set the elasticity to be equal to $\nu = 0.37$, consistent with the estimation results of [Badarinza et al. \(2023\)](#). The constant term of the matching function $\bar{\chi} = 0.23$ is then chosen to match the empirically observed maximum sales probability as a

Figure 5
 “Hockey stick” pattern and concave demand: Model vs. data

The left hand side graph reports the listing premium (difference between listing price and hedonic value) as a function of potential gains (difference between hedonic value and purchase price) in the data as well as for the full and frictionless model versions. The right hand side graph reports the transaction probability within six months of listing in the data and for both model versions. The bottom graph reports the listing premium as a function of potential home equity.



function of the listing price (which is equal to 36%, for a listing premium equal to -9%), for an average level of market tightness in the search data equal to 3.23.

We use the number of residential addresses in the Ordnance Survey address data as a proxy for the stock of residential properties, and find that 2.5% of the housing stock is listed for sale during every semi-annual time period. In addition to being a target in structural estimation, the average listing probability and the steady-state level of market tightness imply a steady-state value of demand (i.e., the number of potential buyers relative to the housing stock) equal to $N_B = 0.08 = 0.025 \times 3.23$.

Finally, to calculate the rate of buyer stamp duty, we consider all realized housing transactions lodged with the U.K. Land Registry during the time period covered by our data (2010–2022). We compute the stamp duty rate that applies to each transaction, given the prevailing tax regime at the time at which it takes place. The representative rate that we use in the calibration ($\tau^b = 0.012$) is given by the average of all stamp duty rates paid over the sample period. For the case of council tax, the situation is complicated by the fact that the taxable base is not the current market value of the property. Depending on the location, the reference price band that applies to each property can refer to market valuations as of 1991, 2003, or the year of construction. As we do not have access to price band data at the property level for each local council, we exploit information about the composition of aggregate government revenue, which suggests that council taxes account for three times as much tax income relative to stamp duty. This ratio allows us to back out the average rate of council tax. We set $\tau^h = 0.012$, which gives the utility cost out of consumption. However, we interpret the tax as a property tax, which is paid as a percentage of the steady state property value. Therefore, for purposes of calculating the government revenues in section 4.7 we scale this by the semi-annual rental yield, set to $y = 2\%$, to produce the relevant property tax.¹⁶

5.2 Structural estimation

The remaining structural parameters are estimated by minimizing a quadratic loss function, where the moments $\mathbf{M}_m(\mathbf{x})$ are calculated numerically from the equilibrium solution of model, for any given set of parameters $\mathbf{x}$. The corresponding empirical

¹⁶Note that in the homeowner’s problem, the role of τ^h is not separately identified from the utility component u . In structural estimation, we therefore vary the net housing utility $u - \tau^h$ to match the targeted moments at the micro level.

Table 2
External calibration

The table reports the model parameters and their sources for externally calibrated parameters as well as for the baseline policy parameters considered in our counterfactual exercises.

Parameter		Value	Target/Source
<i>Externally calibrated parameters</i>			
Time discount	β	0.99	Semi-annual calibration
Housing Stock	N_H	1.00	Normalization
Matching function elasticity	ν	0.37	Listings and search data
Matching function constant	$\bar{\chi}$	0.23	Demand schedule
Number of buyers	N_B	0.08	Market tightness
Listing cost	ϕ	0.01	Hassle cost
Rental yield	y	0.020	Annual 4% rental yield
<i>Policy parameters</i>			
Down payment constraint threshold	γ	0.30	Mortgage interest rate schedule
Stamp duty	τ^b	0.012	Average rate of stamp duty
Council tax	τ^h	0.012	Share of tax revenue

target moments are denoted by $\mathbf{M}_d$:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} (\mathbf{M}_m(\mathbf{x}) - \mathbf{M}_d)' \mathbf{W} (\mathbf{M}_m(\mathbf{x}) - \mathbf{M}_d).$$

We start with a version of the model where we restrict $\eta = \mu = 0$, and estimate six structural parameters, listed in Panel A of Table 3. The estimation targets four empirical moments: the normalized average price level equal to 1, the average listing premium of 8.3%, the demand schedule for a listing premium of 20%, and an average per-period listing probability of 2.5%. The weighting matrix $\mathbf{W}$ is chosen to normalize all empirical target moments to 1. In Panel B of Table 3, we report estimation results from an unrestricted version of the model, where we allow for financial constraints, in addition to reference dependence and loss aversion, which we pin down through the two additional behavioral parameters $\eta > 0$ and $\lambda > 1$ and through the downpayment penalty parameter $\mu > 0$. The additional moments used in estimation are the slopes of the listing premium with respect to nominal potential gains and home equity, measured in the regression of Table A.10.

The estimated coefficients have economically plausible magnitudes, which are

broadly similar between the restricted and unrestricted versions of the model. In the restricted model, the flow utility parameter u is equal to 0.024, consistent with a 2.4% additional flow utility associated with being a homeowner relative to renting. As we normalize the average log house price to be equal to one in steady state, all model quantities should be interpreted in units of the average price. The standard deviations of taste shocks and mobility shocks are robustly 0.3 and 11.9, respectively.

To understand the magnitudes of taste shocks, consider the fact that they accrue over the entire holding period of the asset. In the restricted version of the model, the average estimated value of the taste shock suggests a discounted utility benefit of 141% of the average house price over the expected holding period of 14.7 years in the data. At the same time, the distribution of mobility shocks lies mostly in the negative domain, matching the fact that only 2.5% of the housing stock is listed for sale during each half-year period.

Turning to the unrestricted version of the model, we find that the inclusion of financial constraints and reference dependence increases the magnitude of the mobility shock process that is necessary to match the data, while it decreases the magnitude of the taste shocks. This is driven by the fact that both financial and behavioral frictions decrease the likelihood that properties will be listed for sale, and therefore a higher mobility shock is necessary to match the listing probability in the data. In addition, conditional on listing, most sellers (in particular, those that can realize a positive nominal gain) will be relatively less aggressive than in the frictionless case, as they have a more pronounced incentive to realize the transaction and actually move. This means that a lower value of the buyer’s taste shock is necessary to match the average listing’s observed probability of sale.

Relative to previous studies (Andersen et al., 2022), the estimated magnitude of reference dependence is similar ($\eta = 0.51$), and the strength of loss aversion is higher ($\lambda = 3.46$).¹⁷

¹⁷As we discuss in more detail below, in the current draft we have a reasonable but imperfect match to micro moments that identify the strength of the behavioral preference motive. In future drafts, we expect to improve the fit along these dimensions through estimation algorithms that leverage additional computing power. For estimation, we use the differential evolution global search algorithm. Currently, the estimation of the restricted model uses an order of magnitude more evaluations and estimation iterations to find a proper targeted match than the estimation of the unrestricted model, which is substantially slower to solve.

Table 3
Structural estimation

Panel A
Restricted version

Parameter		Value
<i>Steady-state</i>		
Flow utility	u	0.024
Taste shock (mean)	μ_ε	1.41
Taste shock (st. dev.)	σ_ε	0.30
Mobility shock (mean)	μ_θ	-24.15
Mobility shock (st. dev.)	σ_θ	11.92
Listing cost	ϕ	0.014

Panel B
Unrestricted version

Parameter	Name	Value
<i>Estimated parameters</i>		
Flow utility	u	0.034
Taste shock (mean)	μ_ε	1.16
Taste shock (st. dev.)	σ_ε	0.60
Mobility shock (mean)	μ_θ	-21.62
Mobility shock (st. dev.)	σ_θ	10.49
Reference dependence	η	0.51
Loss aversion	λ	3.46
Downpayment penalty	μ	5.20
Listing cost	ϕ	0.020

5.3 Model fit at the micro level

Table 4 compares the model-implied moments with their counterparts in the data, for both versions of the structural estimation. The restricted model provides a tight fit to the four targeted moments of average price, demand at 20% listing premium, listing probability, and mean listing premium. The unrestricted model can also match all of these four moments relatively well, although the fit especially for the demand schedule is slightly worse than that of the restricted model.

On the other hand, the restricted model cannot explain the listing premium profile for different levels of potential gains or home equity, whereas the unrestricted model provides qualitatively correct behavior for all the four “hockey stick” moments and can match the home equity moments even quantitatively. These are documented in the four bottom rows of Table 4, which show the empirical fit of the unrestricted version of the model in terms of the “hockey stick” profile of the listing premium schedule. For potential gains the model-implied hockey stick right slope has quantitatively correct order of magnitude, but the left slope is flatter than in the data.

Table 4
Targeted moments

Moment	Data	Model version	
		Restricted	Unrestricted
Price (normalization)	1.00	0.99	1.01
Demand at $\ell = 0\%$	0.30	0.29	0.22
Listing probability	0.025	0.025	0.025
Listing premium (mean)	0.083	0.084	0.093
Home equity premium (left slope)	-0.13	0.0	-0.14
Home equity premium (right slope)	-0.098	0.0	-0.092
Potential gain premium (left slope)	-1.08	0.0	-0.29
Potential gain premium (right slope)	-0.36	0.0	-0.27

In Figure 5 we show the model fit for the unconditional listing premium profiles along the potential gain and home equity dimensions, as well as for the demand schedule. The restricted model’s fit of the demand curve is very tight, but this version of the model cannot match the hockey stick listing premium profile at all. The unrestricted model (i.e., the version which includes reference dependence, loss aversion and financial constraints) fits the listing premium profile relatively well, but at the expense of having a slightly worse fit for the demand schedule. The current constellation of structural parameters also has a relatively weak ability to match unconditional listing premia deeper into the negative home equity domain.

6 Results at the aggregate level

6.1 Loss share as sufficient statistic for transaction volumes

We begin by documenting that the loss share measure, i.e. fraction of homeowners who are facing paper losses, can be considered a structurally informed sufficient statistic for explaining transaction volumes. We first discuss how this arises from our model and then demonstrate its explanatory power in the data. Importantly, we show that the loss share measure explains the observed positive price–volume relation in the data: the price growth–volume correlation is not statistically significant once controlling for loss share, whose explanatory power for explaining transaction volumes (as measured in the regression R^2) significantly exceeds that of price growth rates.

6.1.1 Loss shares and transaction volume in the model

Let $s_{it}^* = s^*(r_i, m_{it}, \theta_{it})$ denote the binary indicator describing the optimal extensive margin decision for a homeowner i . The model-implied transaction volume is now given by:

$$\begin{aligned} \text{Transaction Volume} \\ = \int_{r_i, m_{it}} \int_{-\infty}^{\infty} s^*(r_i, m_{it}, \theta_{it}) \alpha(l_{it}^*(r_i, m_{it}, \theta_{it})) d\Phi_{\theta}(\theta_{it}) dF(r_i, m_{it}). \end{aligned} \quad (13)$$

Denote by $\bar{\ell}_{jt}^* \equiv p(\bar{l}_{jt}^*) - \mathbb{E}_t[V_{t+1}^h(r_j, m_{j,t+1})]$ the listing premium as observed by the seller. Using second order approximation for the demand function as $\alpha(\ell) = \alpha_o - \alpha_1 \ell - \alpha_2 \ell^2$, and using a two-bin Riemann sum approximation for the outer integral yields

$$\begin{aligned} \text{Transaction Probability} &= \frac{\text{Transaction Volume}}{\text{Listing Volume}} \\ &\approx \alpha_o - \alpha_1 \left[\text{Loss share} \times \bar{\ell}_{\text{loss}}^* + (1 - \text{Loss share}) \times \bar{\ell}_{\text{gain}}^* \right] \\ &\quad - \alpha_2 \left[\text{Loss share} \times (\bar{\ell}_{\text{loss}}^*)^2 + (1 - \text{Loss share}) \times (\bar{\ell}_{\text{gain}}^*)^2 \right], \end{aligned} \quad (14)$$

where $\bar{\ell}_{\text{loss}}^*$ and $\bar{\ell}_{\text{gain}}^*$ are the conditional average listing premia for homeowners facing losses and gains, respectively. In particular, this shows that conditional on sellers'

listing premium and buyers' demand profiles, the loss share is a simple structurally informed statistic for predicting the transaction probability. Appendix A.7 provides the details of the derivation of this result.

To evaluate the performance of this approximation, in Figure A.9 we consider how the loss share correlates with the true model transaction volumes in a partial equilibrium exercise shifting the potential gain distribution across the gains domain. To gauge the effect of the behavioral mechanism, we constrain the analysis in the model with no mortgage constraints ($\mu = 0$). We feed in the true empirical distribution of potential gains and consider the number of transactions such a distribution generates from the model. In section 6.2 and in Figures 7 and 8 we will later extend this consideration to the full model with mortgage constraints, feeding in the true empirical joint distributions of potential gains and home equities.

In Panel A of Figure A.9 we shift the distribution of potential gains by various amounts by shifting the reference price r_i for all homeowners in the distribution, while holding constant the (general equilibrium) steady-state policy functions for all agents. These shifts generate variation in loss shares, as shown in the horizontal axis in the figure. The vertical axis reports the transaction probability (transaction volume divided by listing volume) for each such shifted distribution in a partial equilibrium exercise, where all agents use the same (general equilibrium) policy functions throughout the exercise.

The loss share statistic explains the transaction probability with $R^2 = 0.95$. Moreover, when setting the binning threshold at 20% instead of 0% gains, consistent with the hockey stick kink being at approximately 20% into the gains domain in this particular model calibration ($\mu = 0$), the explanatory power increases to $R^2 = 0.99$. This is depicted in Panel B of Figure A.9 together with the modeled hockey stick listing premium profile. Thus, merely a two-bin Riemann sum approximation with linear approximation for the demand schedule is able to explain almost all the variation in model-implied volumes across shifted versions of the potential gain distribution. These implications of equation (14) are consistent with the empirical evidence in Figure 1 and the model predictions discussed later in section 6.2 and Figure 6 that transaction fluctuations are driven by the intensive margin (transaction probability) instead of the extensive margin (listing volume).

6.1.2 Loss shares and transaction volumes in the data

As discussed above, the loss share is a strong predictor of transaction volumes in our model. With the intuition provided by our model, we augment the analysis of section 3 by explicitly considering the role of loss share in the price–volume relation. We consider the price–volume regression (2), but add time-varying controls for additional terms from (14) (i.e., loss share and listing premium, with interaction terms). We report the results in Table 5.

The first column reports a price–volume elasticity of 1.69, but once we control for the loss share in the third column, the coefficient on the price growth rate is closer to zero and not statistically significant. Moreover, the explanatory power as measured in regression R^2 is much higher for loss shares. We also consider an additional bin with 20% threshold for potential gains, but the increase in R^2 for doing so is marginal, as documented in the fourth column of the table.

According to equation (13), we would expect listing premium to be another major driver of transaction volumes. We test for this in the fifth column of Table 5. Indeed, the coefficient on listing premium is statistically significant. However, when including the loss share bins in columns (6) and (7) of the table, the coefficient on the listing premium (average and conditional on losses) lose much of their statistical significance and the model’s explanatory power is only marginally improved by adding these additional variables. The price growth rate is again knocked out of the regression. When interacting the loss share measure with the listing premium in column (8), as predicted by equation (14), the interaction terms are statistically significant.

Table 5
Transaction volumes: Explanatory power of alternative model specifications

The table reports the results for a monthly panel regression of log transaction volume on contemporaneous price growth rates across ITL2 regions with regional controls for average listing premium, loss share, and their interaction for new listings measured at monthly frequency. Transaction volumes are deseasonalized by regressing the raw log volumes on quarter fixed effects and taking the residuals. Regional price growth rates are similarly deseasonalized year-on-year growth rates observed in a time fixed-effect of a hedonic model controlling for observable characteristics in realized transactions. Standard errors are clustered on ITL2 level.

	<i>Dependent variable: Transaction volume</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Price growth	1.69*** (0.411)		0.401 (0.232)		0.647** (0.227)	0.374 (0.215)	0.303 (0.194)	
Loss share		-2.19*** (0.078)	-2.07*** (0.073)	-2.32*** (0.108)		-1.64*** (0.246)	-1.76*** (0.242)	-4.58*** (0.686)
Share of gains in [0%, 20%]				2.44*** (0.342)			2.39*** (0.341)	2.05*** (0.361)
Listing premium (mean)					-3.91*** (0.251)	-1.26* (0.565)	-0.934* (0.407)	-1.98*** (0.543)
Cond. listing premium (loss domain)					1.50*** (0.146)	0.349 (0.204)	0.509** (0.173)	-0.026 (0.509)
Cond. listing premium squared								0.330 (0.391)
Loss share × Cond. listing premium (loss domain)								14.6*** (3.58)
Loss share × Cond. listing premium squared (loss domain)								-11.8** (3.63)
ITL2 FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,464	5,460	5,429	5,460	5,429	5,429	5,429	5,460
R ²	0.209	0.439	0.440	0.499	0.408	0.444	0.502	0.519

However, again the explanatory power does not notably differ from only including the paper gains measured in column (4); the loss share in column (2) alone explains 44% of the variation in volumes.¹⁸ Finally, Figure A.6 validates our notion that the loss share provides a sufficient statistic to characterize the aggregate effects of the distribution of nominal returns in the population. As we discretize this distribution into more than two bins (i.e., further distinguishing between losses and gains as above) and compute measure of exposure at the local level, our ability to match the observed variation of transaction volumes only increases marginally.

6.2 Untargeted aggregate moments

We now discuss the model’s ability to explain the data patterns discussed in section 3. We first provide a qualitative discussion of the model mechanisms and then use the calibrated version of the model to evaluate the data patterns discussed in section 3. We do so by considering the impact in the model of potential gain shocks, which shift the potential gain distribution in partial equilibrium.

We note that in the model, the potential gain distribution is endogenous. Therefore, to consider partial equilibrium shifts in this distribution, we consider exogenously specified subpopulations of the endogenous aggregate distribution, matching these subpopulations with the empirical joint distributions of potential gains and home equity at the ITL2 level. We consider the reactions of transaction volumes for these subsets of agents when their potential gains shift. We accomplish these shifts in the model by counterfactually reducing agents’ reference prices. This generates a movement in both the potential gain and home equity distributions, as if the agents had experienced slightly higher price growth (i.e. capital gains) during their holding period.

In all of these counterfactual exercises, we solve for the calibrated model steady state and the associated general equilibrium policy functions for all agents. The steady state policy functions inform us about the agents’ optimal responses to idiosyncratic shocks, which do not affect the aggregate economy. Aggregating over these responses for a set of agents gives us the aggregate partial equilibrium response, when the general equilibrium quantities (market tightness, listing price distribution)

¹⁸Figure A.5 and Table A.6 in the online appendix show that these results remain robust when re-estimated in a sample of properties for which both the buyer and the seller do not have any mortgage contract outstanding.

are kept constant. This differs from having a partial equilibrium model in that the agents still have a highly sophisticated understanding of the general equilibrium to the extent that it affects their steady state policy functions. The simple assumption in these counterfactuals is that these agents confuse an aggregate shock with an idiosyncratic shock, and that they do not account for the shocks’ effects on the behavior of other market participants.

Finally, we note that the restricted model with $\eta = \mu = 0$ has zero response to any potential gain shocks by construction, so a zero response serves as a convenient null in these counterfactuals. Reference prices do not enter the problem in the restricted model, so in that setup, they cannot have an effect on any model quantities.

6.2.1 Positive price-volume comovement

In the model, consider the optimal pricing strategy of the seller, which is given by a non-linear pricing schedule that depends on the location of the reference point. A higher accumulated potential gain makes the seller list for a lower price, in an attempt to attract more buyers and ensure that this gain will be realized. This creates a negative relationship between the level of the potential gain that the seller stands to make and the listing premium that they set. Conversely, if price appreciation since purchase is small or negative, loss-averse sellers want to avoid the possibility of realizing a loss, so they have an incentive to “fish” for higher prices. This behavior is summarized in Figure 5, showing that the model-predicted optimal listing premium matches the shape observed in the data only when sellers are both reference dependent and loss-averse.

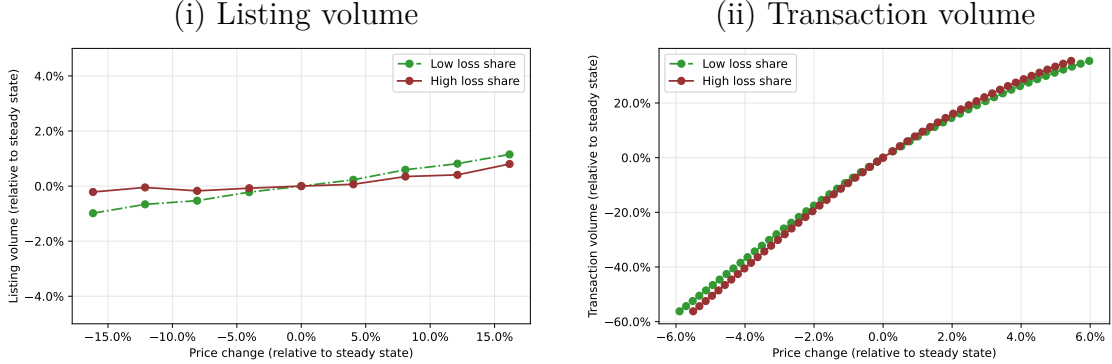
Consider now the effect of a positive valuation shock (i.e., a positive shock in ϵ_u), which increases property values relative to all reference points. This leads all sellers to be less “aggressive,” i.e., to list with lower listing premia, prioritizing lower times on the market. These lower listing premia lead to an increased buyer likelihood of accepting sellers’ offers, which in turn leads to more transactions. This mechanism generates an observed positive correlation between aggregate prices and realized transaction volumes, consistent with the results reported in Panel A of Figure 1.

6.2.2 Asymmetric responses and the role of “paper gains”

While the size of the volume response depends on the mass of homeowners affected by a change in property values, it also depends on the direction of the change in

Figure 6
Model-implied decomposition of the price-volume correlation

The figure reports how listing (left) and transaction volumes (right) response to partial equilibrium shifts in the potential gains distribution for high (90th percentile) and low (10th percentile) loss share distributions. The quantitative magnitudes of the elasticities are not comparable to similar data objects in Figure 1, since the partial equilibrium price shock is different from the shocks faced by the full economy.



property values. Given the asymmetry in the utility function from loss aversion and the mortgage constraint, the model predicts that positive and negative shocks affecting house values create very different volume responses. The results reported in Figure 1 are consistent with this prediction.

We use the calibrated version of the model to demonstrate its ability to deliver these predictions. In Figure 6, we feed in shocks of different sizes and consider the partial equilibrium responses for both listing and transaction volumes. We do this exercise for low loss share (10th percentile) and high loss share (90th percentile) distributions. Similarly to the data patterns documented in Figure 1, the listing volume response is muted, whereas transaction volumes conditional on listing reflect the two key qualitative empirical observations documented in Section 3: (i) the volume response is asymmetric, with volumes responding more to negative than positive shocks, (ii) the responses of volumes are higher for both positive and negative shocks to prices in regions with higher loss shares.

We caveat these results by mentioning that quantitatively, the reported elasticities in this figure are an order of magnitude smaller than those reported in Figure 1. This is due to the nature of the partial equilibrium counterfactuals that we conduct with the model, since here the price-volume relation is only due to the actions of agents facing nominal losses. In general, in response to demand shocks, other agents

would also respond, which we do not consider in this exercise.

Because the model-implied listing premium responds non-linearly to potential gains, the strength of the mechanism described above depends on the aggregate distribution of reference prices prevailing at any point in time. For example, consider a case in which the housing market has appreciated substantially and consistently over a number of periods. Here, all homeowners will have positive nominal gains, the slope of the listing premium will be relatively flat, and the response of volumes to a price shock will be negligible. In contrast, when house values have declined for some period of time, many homeowners face the possibility of realizing losses and the listing premium slope will be very steep, and a change in value will lead to a dramatic adjustment of listing prices and significant variation in observed transaction volumes. Thus, a central prediction of the model is that the share of sellers who face nominal losses is a key aggregate statistic to explain the overall response of transaction volumes to valuation shocks, emphasizing the importance of computing this variable in the data.

In Panel A of Figure 7 we report the relation between the loss shares and price-volume relations across ITL2 regions. We observe that the model implied volume elasticity to prices varies strongly with the loss share. This demonstrates the model's ability to explain the cross-sectional variation in price-volume relationship as a function of the loss share. Furthermore, in the model the loss share is an almost perfect predictor of the price-volume elasticity across sub-populations (mapped to regions in the data).

6.2.3 Extensive vs. intensive margin

The effects described above occur along the intensive margin, i.e., they are driven by sellers' optimal listing price setting and buyers' willingness-to-pay when encountering property sale listings. In parallel with these responses, the model implies that changes in valuation also have an impact on the extensive margin, i.e., when faced with the possibility of realizing a loss, homeowners also have a lower incentive to list the property for sale.

In the model as in the data, the response of listing volumes to changes in house value is more muted because it is partially absorbed by the optimal choice of the listing premium. This difference can be seen by comparing the left and right hand side figures in Figures 6.

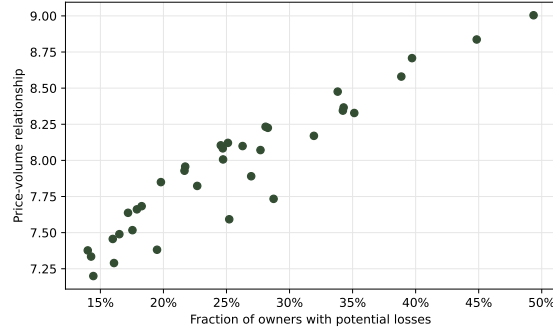
Figure 7

Model-implied cross-regional variation in the price-volume relationship

Panel A plots the model-implied normalized price-volume elasticities against fraction of homeowners facing nominal losses (“loss share”). To produce the loss share measure we match the empirical joint distribution of potential gains and home equities with the corresponding model objects across ITL2 regions, and calculate the model implied loss share for each such distribution. To produce the price-volume elasticity on the vertical axis, we consider a partial equilibrium shift in the distributions, increasing capital gains for everyone and report the elasticity of the volume response. Panel B decomposes the transaction volume responses to listing (i) the listing volume, and (ii) the ratio between the transaction volume and the listing volume, i.e., the transaction probability, similarly to Panel B of Figure 2.

Panel A

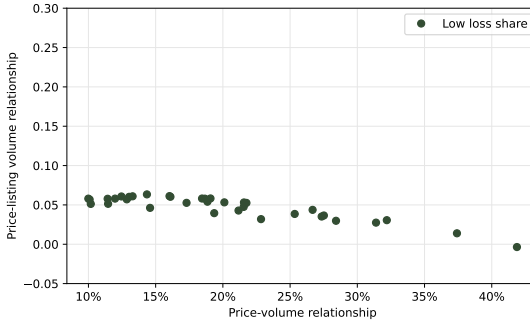
The role of the local loss share



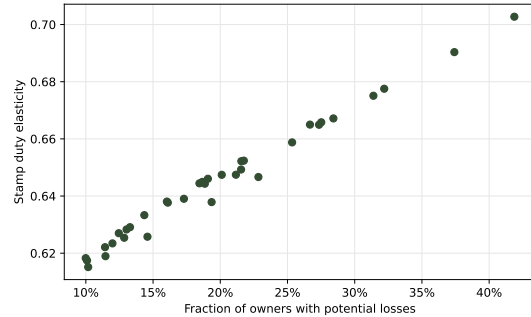
Panel B

Decomposition

(a) Listing volume



(b) Transaction probability



The degree to which this absorption occurs depends on the cost associated with listing (ϕ in the model). If putting a property up for sale incurs a zero cost, all homeowners with a positive realization of the mobility shock (θ) list their properties for sale, and the magnitude of the potential gain just determines how aggressive the “fishing” strategy is. In contrast, when listing costs are very high, e.g., because of estate agent fees, hassle factors, lengthy negotiations with prospective buyers, or

multiple property inspections that are physically invasive, listing only occurs when it is clearly worth paying the high transaction cost; and conditional on listing, sellers are much more conservative, prioritizing a low listing premium and a quick sale.

In Panel B of Figure 7 we decompose the model implied price-volume comovement into its intensive and extensive margin components, similar to Panel B in Figure 2 in the data. Again, the patterns we see in the model closely match those in the data, with the intensive margin explaining the greater part of the cross-sectional variation in the price-volume relation.

6.2.4 Nominal rigidity creates real volatility

Next, consider the response of the market to a negative shock to the value of the property. This opens a gap between the buyers’ willingness-to-pay and the loss-averse sellers’ willingness-to-accept (again, as a consequence of the latter group’s “fish-ing” behavior). Since sellers have the option to defer the transaction and only sell to willing buyers, the realized price distribution will be truncated, only reflecting transactions where relatively higher prices have been achieved. Prices therefore decrease by less than they would in a situation where sellers are not reference-dependent, and volumes decrease by more, absorbing part of the effect of the valuation shock. More generally, for the case of both positive and negative exogenous variation in fundamental quantities, the model predicts that the greater the behavioral lock-in motive, the larger the volatility of volumes, and, all else equal, the lower the volatility of prices.

Thus, our model with reference dependence predicts that in regions with a higher loss share, volumes respond more and prices respond less to fundamental economic shocks. For example, when regions with high nominal loss shares face negative demand shocks, the model predicts that many sellers will be unwilling to reduce prices, which in the data should show up as a dampened price response; correspondingly, the volume response will be amplified. Therefore, the qualitative features embedded in the right hand side of Figure 6 also match the empirical data patterns documented in Figure 3.

7 Policy implications

7.1 Loss share affects responses to tax changes

In our model, an increase in the level of the (flow-based) stamp duty land tax unambiguously affects both prices and transaction volumes. As this tax is paid by the buyer, the effects of a fiscal tightening are similar to a demand shock, and both prices and transaction volumes decrease. Sellers want to avoid the realization of nominal losses, and are less likely to accommodate the buyers' lower willingness-to-pay, trading off time-on-the-market against a higher eventually realized price. At the aggregate level, a higher loss share is therefore associated with a lower price elasticity and a strong response of volumes.

The case of the (stock-based) council tax is similar, albeit subtly different. In a frictionless environment, council tax is non-distortionary, i.e., a higher level reduces property valuations symmetrically for both buyers and sellers, and the number of transactions remains unaffected. However, in the presence of behavioral frictions, sellers have an additional component of utility that prevents them from wanting to realize a potential loss. They therefore maintain aggregate realized prices slightly higher, and the aggregate volume of transactions decreases. As before, the magnitude of this effect depends on the loss share prevailing in each local market.

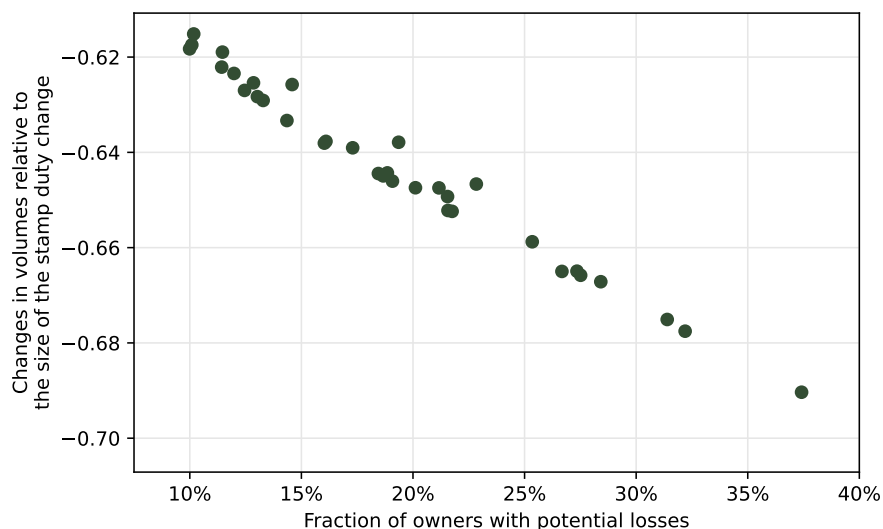
In Figure 8 we show how the model-implied response of volumes to a change in stamp duty depends on the loss share in the location considered. To implement this analysis, we reduce the value of the stamp duty parameter τ^b by 0.01 and consider the partial equilibrium response in transaction volumes. The y-axis of the plot shows the model-implied percentage change in volumes divided by the size of the assumed change in stamp duty. The x-axis shows the loss share at which the policy change is evaluated. The plot reveals that the loss share is an almost perfect predictor of the impact of stamp duty on transaction volumes in this partial equilibrium exercise, which reinforces our interpretation of the loss share as a sufficient statistic to predict housing market responses to stamp duty tax changes.

7.2 Theory vs. Data

Figure A.15 describes the evolution of the tax schedule through time. Before December 2014, the stamp duty rate schedule featured “notches” with taxes discretely increasing at specific threshold price levels. This created strong incentives for buy-

Figure 8
Model-implied volume response for a change in stamp duty

The figure reports the model-implied partial equilibrium percentage response in transaction volumes to a 1pp decrease in the stamp duty rate against the regional loss share for each IT12 region. The volume responses are divided by the change in the stamp duty rate to capture the semi-elasticity of volumes with respect to the stamp duty transaction tax.



ers to seek transactions prices exactly at or below these threshold points to avoid paying a higher rate on the *entire* purchase value of the property, generating visible “bunching” in observed realized price distributions. This system was replaced in December 2014 with a schedule that replaced the discrete increase in the tax rate at the threshold (a “notch”) with a continuous increase in the schedule (a “kink”). This reduced the incentive of buyers to bargain for prices that bunch at particular levels, as the marginal fiscal cost of a transaction price above each threshold is only slightly higher at a kink.

Theoretically, how does seller-side reference dependence influence these outcomes? As discussed above, a seller that faces the possibility of losses when selling at a “notch” threshold transactions price is less likely to accommodate the buyer’s incentive to decrease the price up to the threshold, which should result in an overall lower magnitude of observed bunching in the distribution at the relevant tax threshold. Figure A.14 illustrates this theoretical mechanism.

We validate this mechanism in the data by studying transaction bunching at stamp duty notches. As we describe in appendix section A.11, the tax rate dis-

cretely increases at transaction price thresholds that include £125,000, £250,000 and £500,000. In our validation analysis, we focus on bunching at the £250,000 threshold, which is close to the median house price in the UK.

First, as expected, we document substantial bunching at this price threshold (see appendix, Figure A.15, Panel B). We then estimate excess bunching mass relative to a counterfactual measure of mass at the price threshold. The counterfactual mass is the number of transactions that would occur at each price point if every house were to trade at its hedonic price. Thus, our measure of excess bunching mass is the difference between the number of realised transactions at the tax notch price threshold and the counterfactual mass at the same price point.

Second, we find a negative association between the magnitude of excess bunching mass at the £250,000 price threshold and loss shares across UK ITL2 regions. In Panel (A) of Figure 9 (left), we plot the annual excess bunching mass at this threshold during 2011-2013 against loss shares across UK ITL2 regions. Loss shares are based on the proportion of listings in the loss-domain in a given ITL2 region in the 6-months preceding the period corresponding to the bunching mass. The figure shows a clear negative association, as we confirm using the weighted fit line, and a simple regression analysis whose coefficient we plot in Panel B (left) of the same figure. A higher proportion of listings for properties facing paper losses is associated with lower excess bunching mass, consistent with sellers' reluctance to lower prices.

Third, we find that both the excess bunching mass, and consequently the negative association of the excess bunching mass with loss shares across the ITL2 regions is a lot weaker in periods when the notched tax schedule was modified to a kink in the tax schedule following tax reforms in December 2014. The right hand plot in Panel A of Figure 9 shows this using the annual excess bunching mass and loss shares during 2016-2018. The right hand figure in Panel B shows that the coefficients estimated by regressing the regional excess bunching mass against loss shares in periods with the notched tax schedule yields a statistically significant negative coefficient, and a precisely estimated null effect in periods with a kink in the tax schedule.

7.3 Optimal property tax policy: Laffer curves

As council tax is levied on the entire stock of properties, the total government revenue from this source strictly depends on the average property value at any given point in

Figure 9

Cross-regional variation in the effect of reference dependence on bunching

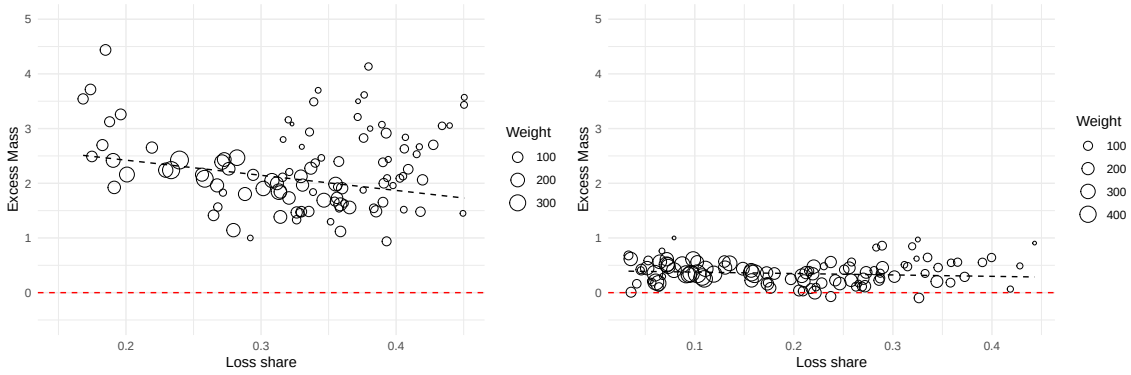
Panel A plots the excess bunching mass for transaction prices in the interval (£245,000, £250,000] against the loss share at the level of ITL2 regions. The figure on the right corresponds to observations in periods with a tax schedule notch; figure on the left corresponds to a tax schedule kink. The dark dotted line shows a linear fit between bunching mass and loss shares weighted by the number of expected transactions for the price band (the weights are shown by the relative size of the circles in both sub-figures). Panel B reports regressions coefficients obtained by regressing excess bunching mass on the loss shares at the ITL2 level for (i) observations during periods when the tax schedule had a notch (2011-2013) at the price £250,000 vs. a kink (2014-2016) and (ii) annual data.

Panel A

The effect of the local loss share on bunching

(a) Notch (2011-2013)

(b) Kink (2016-2018)



Panel B

The effect of the local loss share on bunching using regression coefficients

(a) Across tax schedules:
(Notch vs. Kink)

(b) Annual estimates

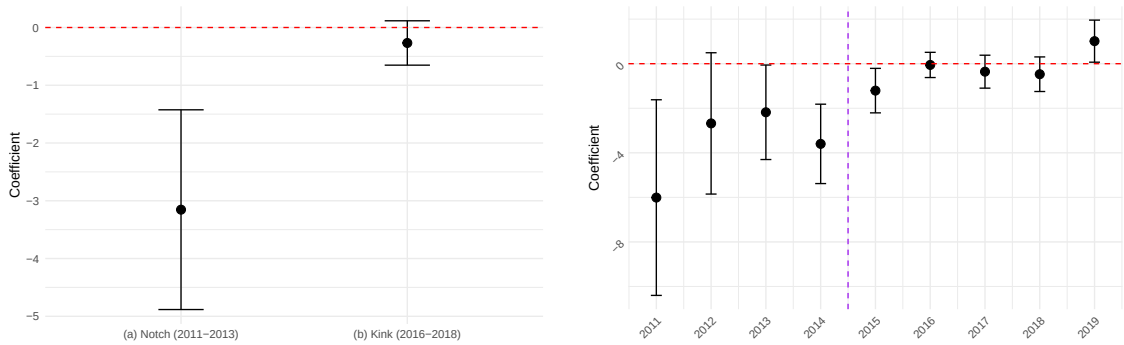
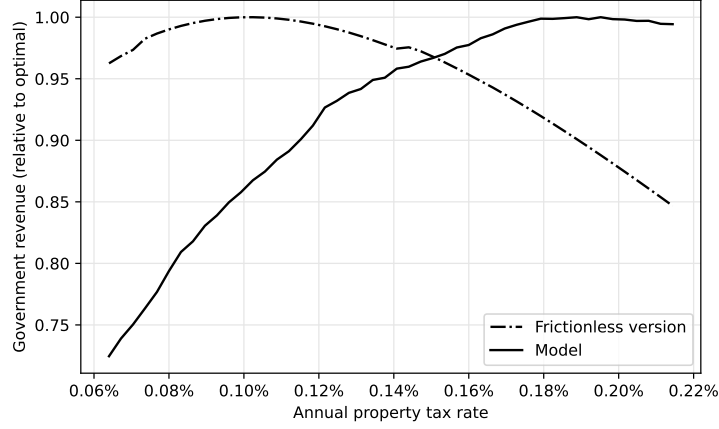


Figure 10
Property tax rate and government revenue

The figure reports the government tax revenues as a function of the property tax rate both for the frictionless model ($\eta = \mu = 0$) and the complete model ($\eta > 0, \lambda > 1, \mu > 0$) versions. Government revenues are scaled by the maximal revenue for each model version separately.



time. As a first-order effect, a higher tax rate decreases prices, which induces a hump-shaped model-implied profile for government revenues as a function of the property tax. While the government might prefer to have high property taxes, increasing the tax rate will eventually produce decreasing government revenues due to the Laffer curve effect. We compare the Laffer curve between the full unrestricted model and the restricted model without behavioral or financial frictions ($\eta = \mu = 0$). Figure 10 plots the government revenue as a function of property tax rate across the different model steady states, each corresponding to a different tax rate. We describe several observations from this figure.

First, note that the tax revenue generated by the unrestricted model (with behavioral and financial frictions) depends on two additional and competing forces compared to the frictionless version of the model. On the one hand, behavioral and financially constrained sellers list at higher prices than non-behavioral and financially unconstrained agents. These higher list prices translate to higher transaction prices, pushing the average transaction price upwards. On the other hand, behavioral and financially constrained agents do not act on all profitable moving opportunities (i.e. high θ shocks) due to their unwillingness to realize a nominal loss on their property. In line with the *rational behavioralism* discussed in section 4.8.2, the buyers foresee this preference of theirs ex ante, when they first purchase the property, and

thus do not value the house as highly as do non-behavioral or financially unconstrained agents. This puts downward pressure on house prices, but this effect is only of second-order, i.e., it affects current decisions through the expectation of future valuation changes. As expected, in our calibration, the first force dominates, and transaction prices in the unrestricted model are higher than in the non-behavioral restricted model, i.e., the tax elasticity is lower—and further decreasing with the magnitude of the loss share.

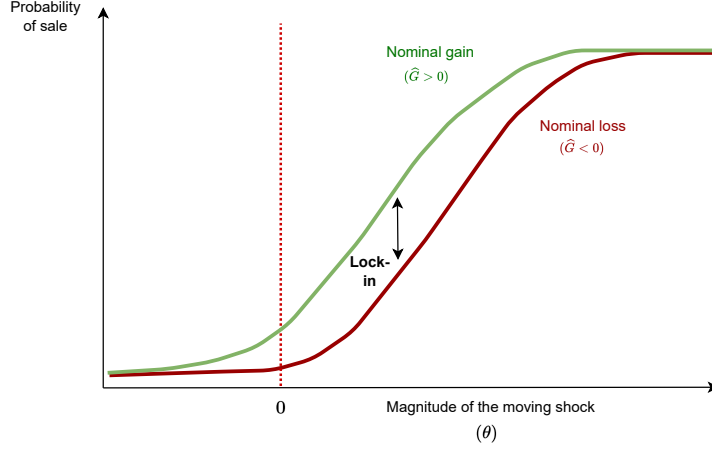
Moreover, in line with the intuition of transaction volumes reacting more strongly than prices, the steady state transaction price is not as sensitive to the property tax in the model with behavioral frictions as it is in the frictionless model. Putting these two effects together, the revenue maximizing tax rate is higher in the unrestricted model and the Laffer curve is shifted to the right compared to the restricted non-behavioral model.

We make one more important observation. While optimal dynamic tax policy is beyond the scope of this paper, we note that an increase in the property tax rate will also generate lock-in during the transition between the steady states. Decreases in nominal prices that arise due to increases in taxes will, *ceteris paribus*, increase the share of households facing nominal losses and therefore affect their behavior as potential sellers. While the government gains revenue from the higher collection of property taxes in the behavioral model, it may lose revenue from other sources (such as, e.g., the labor market), because of reduced income and mobility of behavioral agents. In the case of stamp duty land tax, a similar Laffer-type relationship obtains between the level of the tax and the total government revenue. An increase in the tax level not only reduces prices, but also transaction volumes (which are the taxable base in this case), and the effects continue to scale with the size of the loss share.

7.4 Real effects of nominal shocks

The benefit of the model is that it also allows us to discuss agents' responses to unobservable factors. One important factor of this type is the moving opportunity shock θ . The model predicts that reference-dependent sellers that face the possibility of realizing nominal losses on selling their homes will have dampened responses to positive mobility/productivity shocks. More specifically, reference dependence and loss aversion makes sellers act in such a way as to lower the probability that sale transactions will successfully complete, therefore increasing the chances that they

Figure 11
Mobility shocks and transaction outcomes



miss out on valuable moving opportunities.

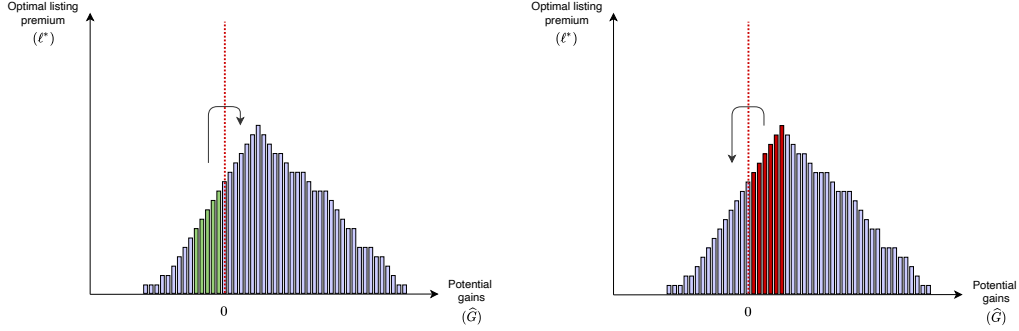
To summarize, the model predicts that the nominal gain/loss situation of the owner will affect their mobility. The diagram in Figure 11 uses the calibrated version of the model to illustrate this effect. We calculate the average probability of sale implied by the model, for different realizations of θ . Sellers in the nominal gain domain are generally more likely to move, but the impact is asymmetric, and most pronounced when the magnitude of the moving opportunity is in an intermediate range.

For low values of θ , the moving probability is low, irrespective of the position of the reference point, because the gains from moving are low. At the opposite end of the distribution, for sufficiently high values of θ , owners have a strong incentive to realize sales regardless of the behavioral motive, meaning that the marginal impact of reference dependence is minimal in this region as well. It is the sellers who face values of θ in an intermediate range that are most likely to fail to realize a sale when facing a nominal loss on selling—they become behaviorally locked in, and reject moving opportunities that would otherwise be valuable. This channel is separate and distinct from both home equity lock-in a la [Bernstein and Struyven \(2022\)](#) and interest-rate lock-in a la [Fonseca and Liu \(2023\)](#).¹⁹

A related implication is that changes in property valuations accompanying fluc-

¹⁹The U.K. mortgage market is dominated by short fixation period ARM instruments ([Badarinza et al., 2018](#)), meaning that the interest-rate lock-in channel is less operative in this market. Despite this, there is a strong similarity in the relationship between prices and volumes between the two markets.

Figure 12
The role of the potential gains distribution



tuating rates of consumer price inflation can lead homeowners to more readily accept or reject moving opportunities (such as. e.g., job offers from other locations), depending on their nominal gain/loss situation. Since labor mobility is a material driver of the level of unemployment ([Head and Lloyd-Ellis, 2012](#)), this generates a link between inflation and the real economy resembling a Phillips-curve relationship, which emerges in the absence of any other source of real rigidity.

7.5 Path dependence

An important feature of the “paper loss” share statistic is that it both encodes the history of past realizations of shocks, and determines the response of the economy in the future. The diagram in [Figure 12](#) illustrates this phenomenon, under the assumption that housing prices are on an upwards trend, and a relatively larger share of the property stock has appreciated since the initial purchase. A shock leading to a property value increase will trigger a volume response primarily because owners that would have seen a modest nominal loss now see a nominal gain, meaning that they are more likely to accept buyer offers. In such an economy, a negative shock to housing values causes a much larger response—triggered by owners that see a potential gain turning into a potential loss—simply because the mass of the latter group of owners is larger than the mass of the former.

Indeed, a more general implication of the model is that after a prolonged period of house price appreciation, the market responds more strongly to a price decrease than to a price increase, because there is more mass in the positive than the negative nominal gain domain; while after a prolonged period of house price depreciation, the market responds more strongly to a price increase, because this triggers a nonlinear

adjustment for a larger share of the housing stock. This also relates to the asymmetric response to shocks discussed in section 6.2.2.

Overall, the path dependence described above is an important feature of our model. A positive valuation shock which increases both prices and volumes leads more owners to reset their reference point to a high level. An eventual negative valuation shock then has a larger impact than in the initial state. Reference dependence therefore generates significant non-linearity in housing market liquidity: The more unusual a “hot market” episode with high liquidity and high prices, the higher the number of owners that lock into a reference point that has a high level, and the deeper the fall in liquidity if valuations later decrease. In terms of policy interventions, the model implies that the transmission of monetary policy to the housing market, for example, will depend to a critical degree on the (recent) history of transactions, with the key conditioning variable being the distribution of reference points prevailing at the time when a policy intervention is rolled out.

8 Conclusions

Careful experiments in a wide variety of economic settings and across a wide range of countries show that agents are reference dependent and loss averse. Such non-standard preferences generate observable behavioral biases in field settings with real stakes—including marathon running, job search, income tax filings, and financial investments (Ingersoll and Jin, 2013; Kleven, 2016; Allen et al., 2017; DellaVigna et al., 2017; Rees-Jones, 2018).

Do these individual biases have a material impact on economic outcomes at the aggregate level? We explore this question in the systemically important residential real estate market. This is also an important market for individual households, as housing is generally the largest asset on the household balance sheet.

We embed loss-averse and reference-dependent sellers into a dynamic equilibrium model of the housing market with search and matching frictions, featuring rich heterogeneity, aggregate shocks, and realistic financial constraints. We structurally estimate the model to fit a rich set of micro data moments. The model delivers a broad set of predictions about (untargeted in structural estimation) aggregate housing market fluctuations. We verify these predictions in the data, confirming the importance of accounting for behavioral frictions to better understand aggregate

outcomes in this important market.

More specifically, we find that the distributions of historically determined nominal reference prices in different regions of the U.K. are tightly empirically linked with the dynamics of aggregate quantities and prices in the housing market. We find that: (i) prices and transaction volumes co-move positively; (ii) the comovement is asymmetric, and more pronounced in locations or times when a high share of houses face “paper losses”; (iii) the fraction of households facing such paper losses is a simple but very important statistic to predict aggregate effects, computed as the share of houses in any given regional market that would realize a nominal loss if it were sold at hedonic value; and (iv) the volatility of housing transaction volumes is substantially more pronounced in regions with high levels of paper losses.

The model has two additional important implications. The first is that nominal shocks affect real outcomes, as homeowners’ moving decisions are directly impacted by their anticipated gains or losses relative to their reference points. The second is that higher property valuations lead new homeowners to set their reference points to high levels, making the housing market more fragile and vulnerable to potential future downturns.

Finally, we validate the importance of these insights by evaluating the appropriate design of housing taxation policies in market populated by behavioral agents. We show that fiscal policy trade-offs in this market alter in the presence of behavioral frictions, and present empirical evidence consistent with this observation. This demonstrates more broadly that behavioral frictions should be taken into account when considering a range of policy interventions targeted at the housing market; we look forward to future work along these lines.

References

- Abel, Andrew B, Janice C Eberly, and Stavros Panageas, “Optimal inattention to the stock market with information costs and transactions costs,” *Econometrica*, 2013, *81* (4), 1455–1481. [2](#)
- Ahn, SeHyoung, Greg Kaplan, Benjamin Moll, Thomas Winberry, and Christian Wolf, “When inequality matters for macro and macro matters for inequality,” *NBER macroeconomics annual*, 2018, *32* (1), 1–75. [5](#), [6](#)
- Aiello, Darren, Jason D. Kotter, and Gregor Schubert, “When Money Moves In: The Consequences of Housing Wealth,” Technical Report, Brigham Young University, Working Paper 2022. [21](#)
- Allcott, Hunt, Benjamin B Lockwood, and Dmitry Taubinsky, “Regressive sin taxes, with an application to the optimal soda tax,” *The Quarterly Journal of Economics*, 2019, *134* (3), 1557–1626. [6](#)
- Allen, Eric J, Patricia M Dechow, Devin G Pope, and George Wu, “Reference-dependent preferences: Evidence from marathon runners,” *Management Science*, 2017, *63* (6), 1657–1672. [2](#), [6](#), [55](#)
- Allen, Jason, Robert Clark, and Jean-François Houde, “Search frictions and market power in negotiated-price markets,” *Journal of Political Economy*, 2019, *127* (4), 1550–1598. [6](#)
- Anagol, Santosh, Benjamin B Lockwood, Allan Davids, and Tarun Ramadorai, “Diffuse Bunching with Frictions: Theory and Estimation,” Technical Report, Working Paper 2022. [6](#), [19](#), [81](#)
- Andersen, Steffen, Cristian Badarinza, Lu Liu, Julie Marx, and Tarun Ramadorai, “Reference dependence in the housing market,” *American Economic Review*, 2022, *112*, 3398–3440. [2](#), [18](#), [23](#), [30](#), [34](#)
- , John Y Campbell, Kasper Meisner Nielsen, and Tarun Ramadorai, “Sources of inaction in household finance: Evidence from the Danish mortgage market,” *American Economic Review*, 2020, *110* (10), 3184–3230. [2](#)

- Anenberg, Elliot and Daniel Ringo**, “The propagation of demand shocks through housing markets,” *American Economic Journal: Macroeconomics*, 2022, *14* (3), 481–507. [6](#)
- **and Patrick Bayer**, “Endogenous Sources Of Volatility In Housing Markets: The Joint Buyer–Seller Problem,” *International Economic Review*, 2020, *61* (3), 1195–1228. [7](#)
- Auerbach, Alan J and Laurence J Kotlikoff**, “Dynamic fiscal policy,” (*No Title*), 1987. [4](#)
- Badarinza, Cristian, John Y Campbell, and Tarun Ramadorai**, “What calls to ARMs? International evidence on interest rates and the choice of adjustable-rate mortgages,” *Management Science*, 2018, *64* (5), 2275–2288. [53](#)
- **, Vimal Balasubramaniam, and Tarun Ramadorai**, “In Search of the Matching Function in the Housing Market,” Technical Report, Working Paper 2023. [30](#)
- Bailey, Martin J, Richard F Muth, and Hugh O Nourse**, “A regression method for real estate price index construction,” *Journal of the American Statistical Association*, 1963, *58* (304), 933–942. [4](#), [12](#)
- Barberis, Nicholas and Wei Xiong**, “Realization utility,” *Journal of Financial Economics*, 2012, *104* (2), 251–271. [4](#)
- Barro, Robert J**, “A theory of monopolistic price adjustment,” *The Review of Economic Studies*, 1972, *39* (1), 17–26. [2](#)
- Bernstein, Asaf and Daan Struyven**, “Housing lock: Dutch evidence on the impact of negative home equity on household mobility,” *American Economic Journal: Economic Policy*, 2022, *14* (3), 1–32. [53](#)
- Bilal, Adrien and Esteban Rossi-Hansberg**, “Location as an Asset,” *Econometrica*, 2021, *89* (5), 2459–2495. [29](#)
- Bracke, Philippe and Silvana Tenreyro**, “History dependence in the housing market,” *American Economic Journal: Macroeconomics*, 2021, *13* (2), 420–443. [2](#)
- Burdett, Kenneth and Kenneth L Judd**, “Equilibrium price dispersion,” *Econometrica: Journal of the Econometric Society*, 1983, pp. 955–969. [6](#)

- Calvo, Guillermo A**, “Staggered prices in a utility-maximizing framework,” *Journal of monetary Economics*, 1983, *12* (3), 383–398. [2](#)
- Chetty, Raj**, “Sufficient statistics for welfare analysis: A bridge between structural and reduced-form methods,” *Annu. Rev. Econ.*, 2009, *1* (1), 451–488. [2](#)
- , “Behavioral economics and public policy: A pragmatic perspective,” *American Economic Review*, 2015, *105* (5), 1–33. [6](#)
- DellaVigna, Stefano, Attila Lindner, Balázs Reizer, and Johannes F Schmieder**, “Reference-dependent job search: Evidence from Hungary,” *The Quarterly Journal of Economics*, 2017, *132* (4), 1969–2018. [2](#), [6](#), [55](#)
- Diamond, Peter A**, *A search-equilibrium approach to the micro foundations of macroeconomics*, Vol. 1982, MIT press, 1984. [6](#)
- Diaz, Antonia and Belén Jerez**, “House prices, sales, and time on the market: A search-theoretic framework,” *International Economic Review*, 2013, *54* (3), 837–872. [6](#)
- Fisher, Jack, Alessandro Gavazza, Lu Liu, Tarun Ramadorai, and Jagdish Tripathy**, “Refinancing cross-subsidies in the mortgage market,” Technical Report, Working Paper 2023. [3](#), [7](#)
- Fonseca, Julia and Lu Liu**, “Mortgage Lock-In, Mobility, and Labor Reallocation,” Technical Report, Working Paper 2023. [3](#), [7](#), [53](#)
- Genesove, David and Christopher Mayer**, “Loss aversion and seller behavior: Evidence from the housing market,” *The quarterly journal of economics*, 2001, *116* (4), 1233–1260. [2](#), [12](#), [30](#), [80](#)
- and **Lu Han**, “Search and matching in the housing market,” *Journal of Urban Economics*, 2012, *72* (1), 31–45. [2](#)
- Grindaker, Morten, Artashes Karapetyan, Espen Moen, and Plamen Nenov**, “Transaction Sequence Decisions and Housing Market Volatility: Evidence from Norway,” Technical Report 2021. [7](#)
- Guren, Adam M**, “House price momentum and strategic complementarity,” *Journal of Political Economy*, 2018, *126* (3), 1172–1218. [6](#), [30](#)

- Han, Lu and William C Strange**, “The microstructure of housing markets: Search, bargaining, and brokerage,” *Handbook of regional and urban economics*, 2015, 5, 813–886. [6](#)
- Head, Allen and Huw Lloyd-Ellis**, “Housing liquidity, mobility, and the labour market,” *Review of Economic Studies*, 2012, 79 (4), 1559–1589. [54](#)
- Huckfeldt, Christopher**, “Understanding the scarring effect of recessions,” *American Economic Review*, 2022, 112 (4), 1273–1310. [6](#)
- Ingersoll, Jonathan E and Lawrence J Jin**, “Realization utility with reference-dependent preferences,” *The Review of Financial Studies*, 2013, 26 (3), 723–767. [2](#), [55](#)
- Kahneman, Daniel and Amos Tversky**, “Prospect Theory: An Analysis of Decision under Risk,” *Econometrica*, 1979, 47 (2), 263–291. [2](#)
- Kleven, Henrik Jacobsen**, “Bunching,” *Annual Review of Economics*, 2016, 8, 435–464. [6](#), [55](#)
- Krainer, John**, “A theory of liquidity in residential real estate markets,” *Journal of Urban Economics*, 2001, 49 (1), 32–53. [6](#)
- Krusell, Per and Anthony A Smith Jr**, “Income and wealth heterogeneity in the macroeconomy,” *Journal of Political Economy*, 1998, 106 (5), 867–896. [5](#), [6](#)
- Liu, Lu**, “The demand for long-term mortgage contracts and the role of collateral,” Technical Report, Working Paper 2022. [11](#), [74](#)
- Lucas, Robert E Jr.**, “Macroeconomic priorities,” *American economic review*, 2003, 93 (1), 1–14. [5](#)
- Mankiw, N Gregory and Ricardo Reis**, “Pervasive stickiness,” *American Economic Review*, 2006, 96 (2), 164–169. [2](#)
- Merlo, Antonio and Francois Ortalo-Magne**, “Bargaining over residential real estate: evidence from England,” *Journal of Urban Economics*, 2004, 56 (2), 192–216. [6](#)

- Moen, Espen R, Plamen T Nenov, and Florian Sniekers**, “Buying first or selling first in housing markets,” *Journal of the European Economic Association*, 2021, 19 (1), 38–81. [7](#)
- Mullainathan, Sendhil, Joshua Schwartzstein, and William J Congdon**, “A reduced-form approach to behavioral public finance,” *Annu. Rev. Econ.*, 2012, 4 (1), 511–540. [6](#)
- Ngai, L Rachel and Silvana Tenreyro**, “Hot and cold seasons in the housing market,” *American Economic Review*, 2014, 104 (12), 3991–4026. [6](#), [12](#)
- O’Donoghue, Ted and Matthew Rabin**, “Optimal sin taxes,” *Journal of Public Economics*, 2006, 90 (10-11), 1825–1849. [6](#)
- Piazzesi, Monika, Martin Schneider, and Johannes Stroebe**, “Segmented housing search,” *American Economic Review*, 2020, 110 (3), 720–59. [6](#)
- Rees-Jones, Alex**, “Quantifying loss-averse tax manipulation,” *The Review of Economic Studies*, 2018, 85 (2), 1251–1278. [2](#), [6](#), [55](#)
- Reis, Ricardo**, “Inattentive consumers,” *Journal of monetary Economics*, 2006, 53 (8), 1761–1800. [2](#)
- , “Inattentive producers,” *The Review of Economic Studies*, 2006, 73 (3), 793–821. [2](#)
- Shiller, Robert J**, “Arithmetic repeat sales price estimators,” *Journal of Housing Economics*, 1991, 1 (1), 110–126. [4](#)
- Stein, Jeremy C**, “Prices and trading volume in the housing market: A model with down-payment effects,” *The Quarterly Journal of Economics*, 1995, 110 (2), 379–406. [4](#), [7](#), [12](#), [23](#), [82](#)
- Taubinsky, Dmitry and Alex Rees-Jones**, “Attention variation and welfare: Theory and evidence from a tax salience experiment,” *The Review of Economic Studies*, 2018, 85 (4), 2462–2496. [6](#)
- Wheaton, William C**, “Vacancy, search, and prices in a housing market matching model,” *Journal of Political Economy*, 1990, 98 (6), 1270–1292. [6](#)

Behavioral Lock-In: Aggregate Effects of Reference Dependence in the Housing Market

ONLINE APPENDIX

Cristian Badarinza, Tarun Ramadorai,
Juhana Siljander, Jagdish Tripathy

August 22, 2024

A Appendix

Contents

A.1 Additional tables and figures	65
A.2 Bunching of transaction prices	80
A.3 Seller optimization: Listing premia	81
A.4 Buyer optimization: Matching outcomes	83
A.5 Notes on the construction of the mortgage data	84
A.6 Listings vs. transaction volumes	85
A.7 Loss share as sufficient statistic: Analytical derivations	86
A.8 Computational appendix	87
A.9 Model calibration	88
A.10 Theoretical intuition: Bunching at tax threshold levels	90
A.11 Housing taxation in the UK	90

List of Figures

A.1	Reference dependence and loss aversion	65
A.2	Reference dependence, loss aversion and financial constraints	66
A.3	Actual price vs. hedonic price	66
A.4	Price-volume correlation in U.S. data	67
A.5	Cross-regional variation in the price-volume relationship: Non-mortgage sample	73
A.6	Loss share as a sufficient statistic	74
A.7	Initial loan-to-value ratios and costs	74
A.8	Amortization and repayment of mortgage loans	75
A.9	Loss share and transaction probability	76
A.10	“Paper losses” and home equity	79
A.11	Regional variation in bunching of transaction prices	80
A.12	“Hockey stick” pattern across regions	82
A.13	Search and matching outcomes	84
A.14	Illustration of bunching at tax kink points	90
A.15	Transaction tax schedule and bunching	92

List of Tables

A.1	Price-volume correlation: International comparison	68
A.2	Transactions and listings data: sample description and cleaning	69
A.3	Total number of observations across mortgage-stock snapshots	70
A.4	Summary statistics on listings sample merged with mortgage data	71
A.5	Price-volume correlation: intensive vs. extensive margins	71
A.6	Robustness check: Restricted sample	72
A.7	Summary statistics on mortgages originated in 2015	72
A.8	Price volume regressions with leads and lags	77
A.9	Loss share and listing premium interaction with leads and lags	78
A.10	Listing premia: Conditional “hockey stick” patterns	83

A.1 Additional tables and figures

Figure A.1
Reference dependence and loss aversion

Plot (i) reports the distribution of differences between the realized price in a given sale transaction and the price for which the property has been initially purchased (the nominal realized gain). Plot (ii) reports the difference between this distribution and a counterfactual version of it, obtained under the assumption that all transactions occur at the estimated hedonic value (excess mass).

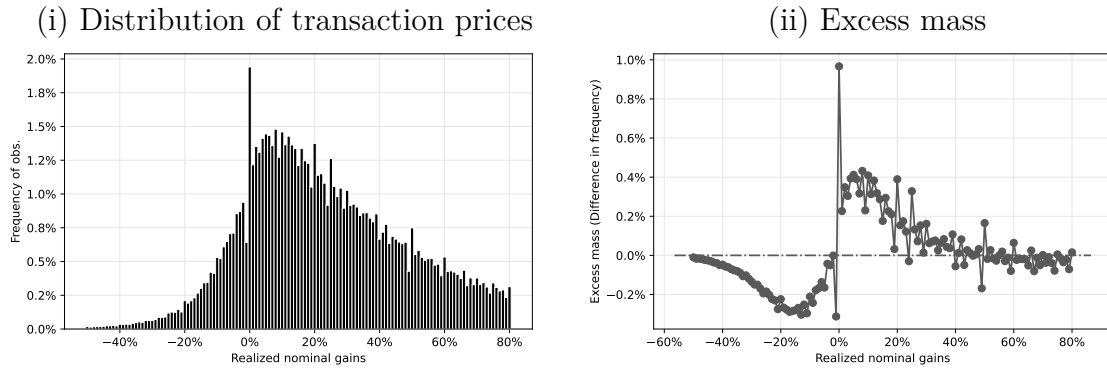


Figure A.2
Reference dependence, loss aversion and financial constraints

The figure plots average listing premia, i.e., the percent differences between the listing price and the hedonic property value at the time of listing, for different values of potential gains. We restrict the horizontal axis, for a more convenient graphical representation of effects around zero potential gains.

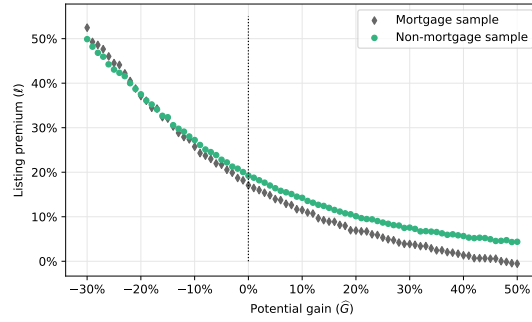


Figure A.3
Actual price vs. hedonic price

The figure illustrates the in-sample fit of our hedonic model. The horizontal axis shows the realized log transaction price, and the vertical axis the predicted log price from the hedonic model. We report a bin-scatter plot based on 20 frequency bins.

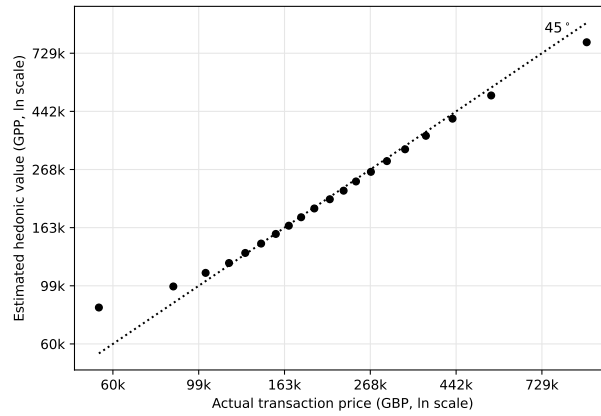


Figure A.4
Price-volume correlation in U.S. data

The figure reports year-over-year changes in housing prices and the correlation with the volume of realized transactions, at the level of U.S. states, for the period between January 2010 and December 2022. We express the volume series as a relative deviation to the location-specific mean, and the price growth as a relative percent difference to the location-specific mean. Error bars indicate 95% confidence intervals for bin-specific means.

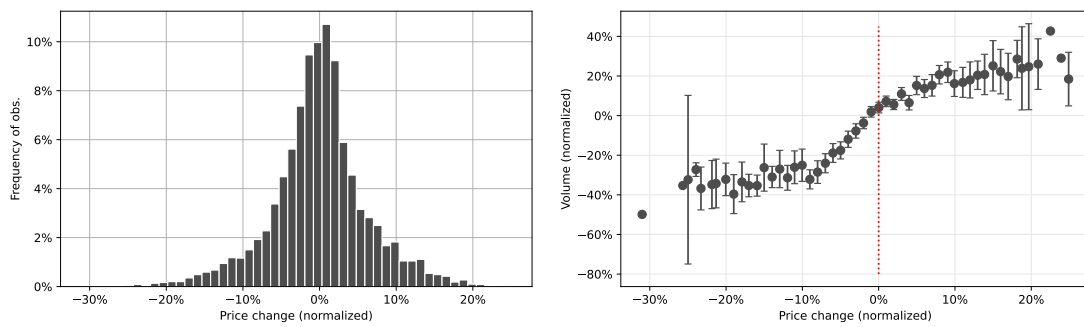


Table A.1
Price-volume correlation: International comparison

The table reports estimated coefficients from the following regression specification:

$$\ln V_{it} = \gamma_i + \beta \Delta \ln P_{it} + \varepsilon_{it},$$

where i is an *ITL2* region, t is a month and Δ is a year-over-year difference operator. *, **, *** indicate statistical significance at the 10%, 5% and 1% level, based on standard errors clustered at the level of ITL2 regions. In the U.K., we decompose the volume effect to listing volume and the realized transaction volume conditional on listing (i.e. transaction probability / conversion rate of listings to transactions).

	U.K.		U.S.
	Listing	Realized	Realized
	volume	volume	volume
Price change (YoY)	0.318***	2.270***	2.42***
	(0.119)	(0.283)	(0.220)
No. of obs.	4,871	4,871	5,927
R ²	0.669	0.301	0.910

Table A.2
Transactions and listings data: sample description and cleaning

Rightmove listings data		HM Land Registry Price Paid data	
All unique listings	21,014,152	All transactions	28,012,267
Ambiguous addresses	-2,019,206	Transactions before 2010	-16,228,143
Not in England or Wales	-908,193	Ambiguous addresses	-964,799
Total	18,086,753		10,819,325
Development prop.	1,130,725		
Non-development prop.	16,956,028		
Listings/transactions match			
Unmatched to a preceding transaction	-10,015,534	Unmatched to a preceding listing	-4,308,913
Development prop.	-1,102,693	Transaction over 365 days after listing archiving	-868,231
Non-development prop.	-8,912,841		
	8,071,219		5,642,181
Development prop.	28,032		
Non-development prop.	8,043,187		
Hedonic sample			
Auction properties	-80,670		-76,766
Transaction within 30 days of first listing	-56,635		
Incomplete hedonic characteristics	-2,141,343		-1,635,494
Final listings data	5,764,539	Hedonic model data	3,929,921

Table A.3

Total number of observations across mortgage-stock snapshots

The table shows the number of observations in the U.K. mortgage stock associated with a property value (column 1), with a unique combination of property value, transaction date and granular postcodes (used to merge to the U.K. land registry, column 2), and with a unique combination of granular postcodes and borrower date of birth (used to track mortgages over snapshots, column 3).

Snapshot	Total	Unique combination for merge	
		with LandReg	across snapshots
2015H1	6,804,852	6,727,027	6,547,065
2015H2	6,903,364	6,819,688	6,625,624
2016H1	6,966,672	6,880,262	6,655,600
2016H2	7,118,232	7,025,876	6,792,460
2017H1	7,195,250	7,098,368	6,854,541
2017H2	7,457,215	7,354,596	7,064,382
2018H1	7,481,944	7,375,001	7,064,379
2018H2	7,517,354	7,410,966	7,052,391
2019H1	7,769,324	7,652,926	7,284,116
2019H2	7,768,736	7,654,581	7,284,871
2020H1	6,897,770	6,811,842	6,485,018
2020H2	8,103,209	7,979,305	7,669,896
2021H1	8,273,139	8,142,959	7,835,923
2021H2	8,121,613	7,990,651	7,649,971
2022H1	8,288,562	8,154,020	7,771,863
2022H2	8,488,874	8,351,212	7,950,613

Table A.4

Summary statistics on listings sample merged with mortgage data

The table shows summary statistics for our listings sample that are merged with the mortgage data based on unique combinations of property value, transaction date and granular postcodes. We further condition the sample to only consider listings that are eventually transacted, and have non-zero home equity close to the listing date.

	mean	sd	p10	p25	p50	p75	p90
Previous price (£)	204,468	138,251	97,000	125,000	170,000	240,000	340,000
Hedonic price (£)	250,533	167,267	110,150	145,200	205,815	302,212	438,074
$\hat{G}$	0.19	0.24	-0.11	0.03	0.18	0.34	0.49
$\hat{H}$	0.42	0.25	0.13	0.27	0.42	0.58	0.74
$\hat{H}$ (Binned)	0.42	0.16	0.21	0.32	0.43	0.53	0.62

Table A.5

Price-volume correlation: intensive vs. extensive margins

The table reports estimated coefficients from the following regression specifications in the Rightmove listings sample:

$$\begin{aligned}\ln(V_t) &= \gamma + \beta P_{\Delta,t} + \varepsilon_t, \\ \ln(L_t) &= \gamma_L + \beta_L P_{\Delta,t} + \varepsilon_{t,V}, \\ \ln(V_t/L_t) &= \gamma_{VL} + \beta_{VL} P_{\Delta,t} + \varepsilon_{t,VL},\end{aligned}$$

where t is a month and $P_{\Delta,t}$ is a de-seasonalised year-over-year price growth index in the U.K.. *, **, *** indicate statistical significance at the 10%, 5% and 1% level.

	Transaction volume $\ln(V_t)$ (β)	Supply of listings $\ln(L_t)$ (β_L)	Conversion rate $\ln(V_t/L_t)$ (β_{VL})
Price growth index ($P_{\Delta,t}$)	3.395*** (1.053)	1.321* (0.745)	2.074*** (0.635)
Number of obs.	132	132	132
R ²	0.074	0.024	0.076

Table A.6
Robustness check: Restricted sample

This table reports the results for a monthly regression of log transaction volume on contemporaneous price growth rates and loss shares at the level of ITL2 regions and months. The analysis uses transaction volumes repeat sales price indices and loss shares computed in the sub-sample of Land Registry transactions for which neither the buyer nor the seller are associated with a mortgage contract. The loss share measure is calculated monthly for properties which have at least one transaction in the data preceding each respective month. The data cover the period between 2010–2022. *, **, *** indicate statistical significance at the 10%, 5% and 1% level, based on standard errors clustered at the level of ITL2 regions.

Log Transaction Volumes (Non-mortgage)			
Price growth (Non-mortgage)	4.252*** (0.3299)	0.8470*** (0.1144)	
Loss share (Non-mortgage)	-1.372*** (0.0388)	-1.393*** (0.0377)	
Fixed effects (ITL2)	Yes	Yes	Yes
No. of obs.	7,105	7,489	7,105
R ²	0.196	0.230	0.442

Table A.7
Summary statistics on mortgages originated in 2015

The table shows summary statistics for mortgages originated in the first half of 2015 (2015H1) that have a non-missing balance in each snapshot until 2022H2. We observe the original loan and subsequent balance in half-yearly snapshots over 7 years for this sample, and use it for the amortization schedule shown in Figure A.8 used to fit the model.

	mean	sd	p10	p25	p50	p75	p90
Original Loan (£)	157,564	128,993	52,813	81,464	127,547	195,725	289,995
Balance (£)	154,386	126,774	51,652	80,008	125,117	191,876	283,556
Interest Rate (%)	2.95	1.06	1.84	2.29	2.79	3.49	4.45
Original Term (months)	471	1,370	156	204	288	360	420
Monthly Payment (£)	766	579	287	429	641	954	1,382
Incentivised (ID)	0.94	0.24	1	1	1	1	1

Figure A.5
Cross-regional variation in the price-volume relationship: Non-mortgage sample

The figure plots regression coefficients from regressions of the realized transaction volume on the year-on-year price change in each ITL2 region, for the period between January 2010 and December 2022. The analysis uses transaction volumes repeat sales price indices and loss shares computed in the sub-sample of Land Registry transactions for which neither the buyer nor the seller are associated with a mortgage contract. The loss share measure is calculated monthly for properties which have at least one transaction in the data preceding each respective month.

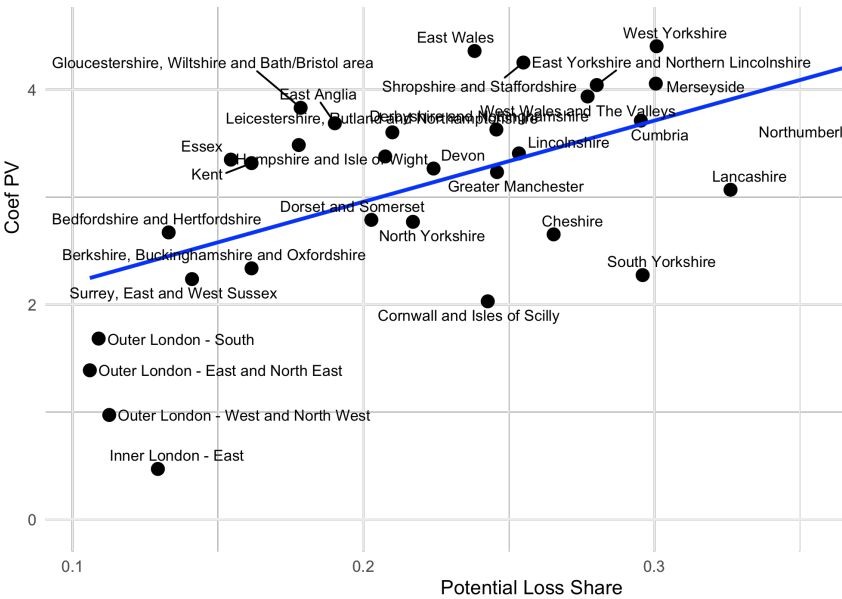


Figure A.6
Loss share as a sufficient statistic

The figure plots cumulative R-square statistics from a regressions of the realized transaction volume on the year-on-year price change and loss share in each ITL2 region, for the period between January 2010 and December 2022, corresponding to table 5. In addition to the loss share, we further discretize the nominal gain distribution in multiple bins (indicated on the horizontal axis), and add to the regression the share of nominal gains in each bin in each ITL2 region and each period.

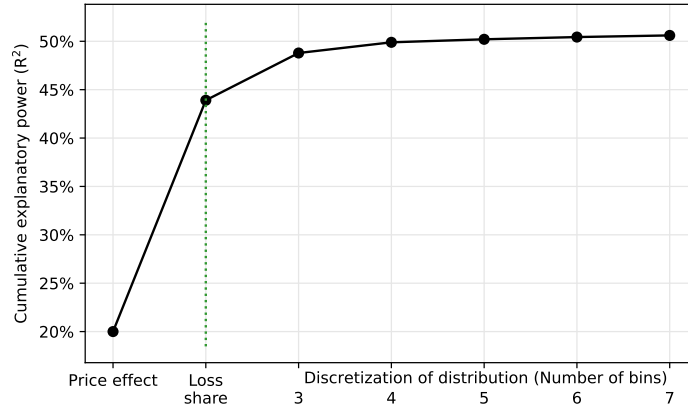


Figure A.7
Initial loan-to-value ratios and costs

The figure reports the distribution of loan-to-value ratios at origination and the mortgage credit interest rate spread paid on 5-year fixed-rate mortgages across loan-to-value bands estimated by Liu (2022) for mortgage contracts issued between 2013 and 2017.

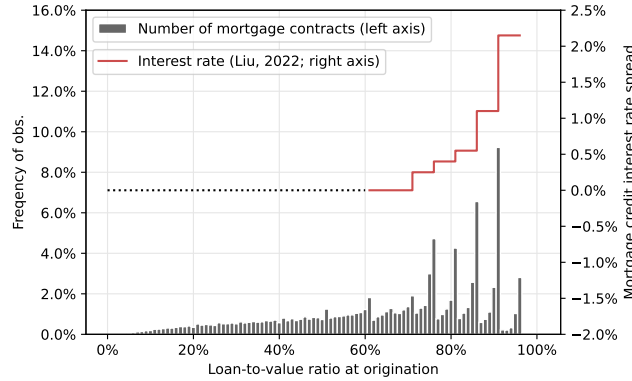


Figure A.8
Amortization and repayment of mortgage loans

The figure illustrates the evolution of outstanding mortgage balances in our data in half-yearly snapshots for a sub-sample of mortgages originated in the first-half of 2015 using regression analysis. The estimated coefficients reflect average amortization rate for half-yearly lags to origination, and based on the following specification:

$$b_{it} = \sum_t \mathbb{D}_t + \varepsilon_{it},$$

where $b_{it} = l_{it}/l_{i,2015H1}$ is based on the outstanding balance for mortgage i in half-yearly snapshot t (l_{it}) and original loan size ($l_{i,2015H1}$), and $\mathbb{D}_t$ are half-yearly dummies.

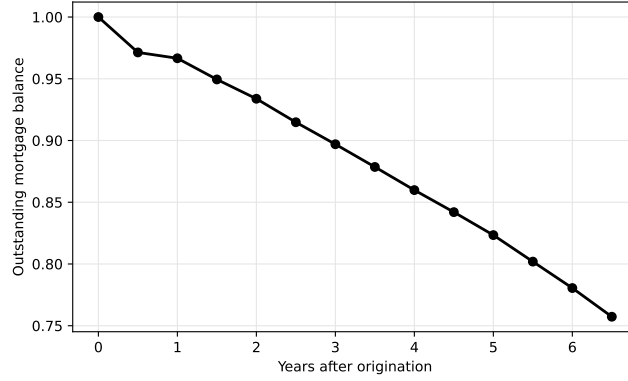
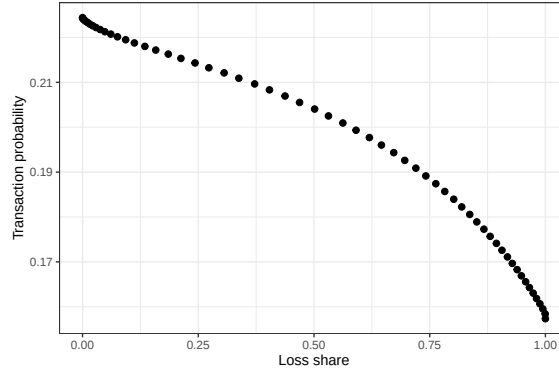


Figure A.9
Loss share and transaction probability

This figure considers the effect of shifting the potential gains distribution on transaction probability. In panel A, the vertical axis shows the transaction probability from the model without mortgage constraints ($\mu = 0$). Each unit of observation corresponds to a partial equilibrium simulation from the model shifting the potential gains distribution by various amounts reflected in the loss share measure in the horizontal axis. In Panel B, on the left we show that the hockey stick kink is not at 0, but at approximately 20% into the gains domain. The right hand side figure of Panel B reports the transaction probability against the share of properties with potential gains less than 20%, demonstrating a linear relationship with $R^2 = 0.99$.

Panel A
Loss share and transaction probability



Panel B
Hockey stick and share of properties with gains < 20%

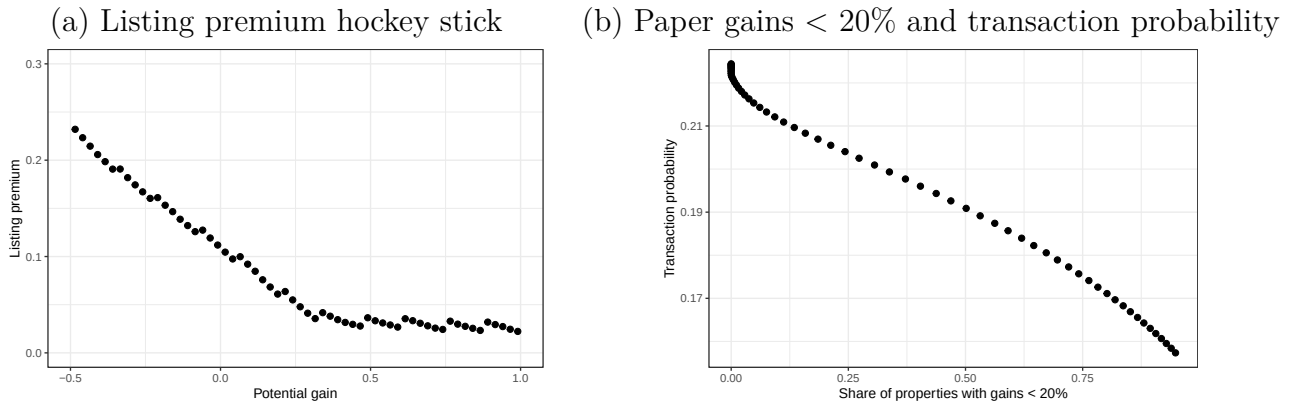


Table A.8
Price volume regressions with leads and lags

The table reports the coefficients and R^2 s for four different regressions regressing (log) transaction volumes on price growth rates, loss share and average listing premium, for 13 different specifications with monthly leads (from -1 to -6) and lags (1 to 6) for the dependent variables.

	<i>Dependent variable: Transaction volume</i>												
	Price growth lag (months)												
	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6
Price growth	1.71 (0.55)	1.94 (0.54)	1.85 (0.54)	2.12*** (0.54)	1.82*** (0.55)	1.64*** (0.57)	1.59** (0.73)	0.92 (0.74)	0.56 (0.74)	0.42 (0.75)	0.39 (0.76)	-0.08 (0.76)	-0.47 (0.77)
R ²	0.06	0.08	0.07	0.09	0.07	0.05	0.03	0.01	0.004	0.002	0.002	0.0001	0.003
Price growth	0.89 (0.54)	1.16 (0.53)	1.19 (0.54)	1.48*** (0.53)	1.30** (0.55)	1.26** (0.57)	1.43** (0.59)	1.03 (0.78)	0.62 (0.78)	0.20 (0.78)	0.32 (0.79)	-0.21 (0.79)	-0.77 (0.79)
Listing premium	-4.12 (0.80)	-3.93 (0.80)	-3.33 (0.82)	-3.30*** (0.80)	-2.79*** (0.83)	-2.29*** (0.85)	-2.06** (0.88)	0.52 (1.16)	0.36 (1.19)	-1.28 (1.23)	-0.47 (1.27)	-0.87 (1.30)	-2.24* (1.31)
R ²	0.20	0.20	0.16	0.19	0.14	0.10	0.10	0.01	0.004	0.01	0.003	0.003	0.02
Price growth	-0.13 (0.50)	0.23 (0.51)	0.20 (0.52)	0.57 (0.52)	0.35 (0.53)	0.32 (0.55)	0.59 (0.59)	0.02 (0.80)	-0.34 (0.81)	-0.43 (0.81)	-0.34 (0.82)	-0.86 (0.82)	-1.28 (0.82)
Loss share	-1.83 (0.21)	-1.70 (0.22)	-1.64 (0.22)	-1.55*** (0.22)	-1.49*** (0.23)	-1.38*** (0.23)	-1.23*** (0.25)	-0.88** (0.34)	-0.89** (0.34)	-0.87** (0.35)	-0.78** (0.35)	-0.84** (0.35)	-0.90** (0.36)
R ²	0.36	0.34	0.32	0.32	0.28	0.24	0.19	0.05	0.05	0.04	0.03	0.04	0.04
Price growth	-0.25 (0.50)	0.13 (0.51)	0.04 (0.51)	0.45 (0.51)	0.18 (0.52)	0.12 (0.54)	0.42 (0.58)	-0.37 (0.76)	-0.73 (0.76)	-0.62 (0.81)	-0.69 (0.81)	-1.17 (0.82)	-1.37 (0.84)
Loss share	-2.48 (0.37)	-2.23 (0.37)	-2.49 (0.37)	-2.24*** (0.38)	-2.44*** (0.38)	-2.49*** (0.39)	-2.19*** (0.42)	-3.05*** (0.55)	-2.95*** (0.56)	-1.76*** (0.60)	-2.09*** (0.61)	-1.97*** (0.61)	-1.22* (0.62)
Listing premium	2.66 (1.23)	2.18 (1.25)	3.48 (1.25)	2.84** (1.26)	3.87*** (1.28)	4.50*** (1.31)	3.94*** (1.42)	8.85*** (1.85)	8.55*** (1.91)	3.75* (2.11)	5.67*** (2.16)	4.98** (2.21)	1.40 (2.27)
R ²	0.38	0.35	0.35	0.34	0.32	0.29	0.23	0.18	0.16	0.06	0.08	0.07	0.05

Table A.9
Loss share and listing premium interaction with leads and lags

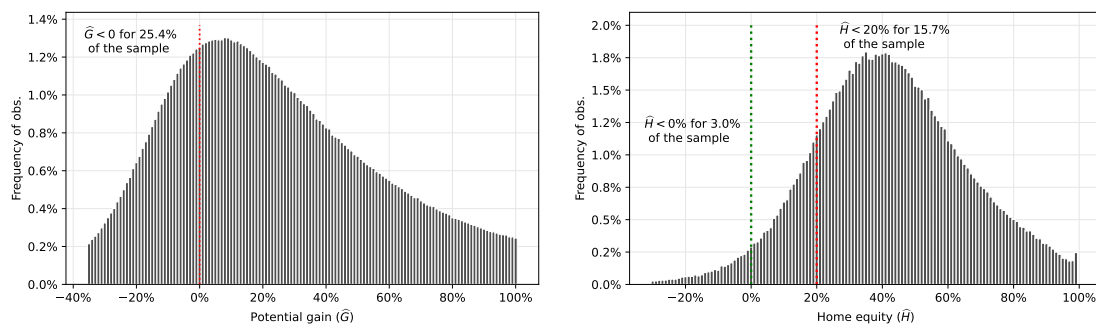
The table reports the coefficients and R^2 s for regression transaction volumes on price growth, loss share, listing premium, and the interaction of the last two for 13 different specifications with monthly leads (from -1 to -6) and lags (1 to 6) for the dependent variables.

	<i>Dependent variable: Transaction volume</i>												
	Price growth lag (months)												
	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6
Price growth	-0.11 (0.49)	0.30 (0.49)	0.23 (0.48)	0.60 (0.50)	0.38 (0.49)	0.35 (0.50)	0.61 (0.55)	-0.03 (0.69)	-0.40 (0.69)	-0.40 (0.75)	-0.68 (0.74)	-1.23 (0.76)	-1.58** (0.80)
Listing premium	6.75 (1.74)	7.22 (1.75)	9.01 (1.73)	7.21*** (1.78)	9.80*** (1.76)	11.28*** (1.77)	9.66*** (1.98)	18.95*** (2.47)	20.20*** (2.60)	16.61*** (3.09)	21.21*** (3.37)	21.05*** (3.81)	16.86*** (4.29)
Loss share	0.97 (1.13)	2.02 (1.13)	2.17 (1.12)	1.44 (1.15)	2.56** (1.14)	3.22*** (1.15)	2.62** (1.28)	5.47*** (1.60)	6.34*** (1.63)	7.53*** (1.82)	8.15*** (1.89)	7.98*** (2.07)	7.95*** (2.28)
Listing premium × Loss share	-24.64 (7.65)	-30.36 (7.69)	-33.30 (7.59)	-26.34*** (7.82)	-35.67*** (7.74)	-40.78*** (7.79)	-34.42*** (8.68)	-60.83*** (10.86)	-66.95*** (11.17)	-68.17*** (12.75)	-76.63*** (13.48)	-75.14*** (15.02)	-69.75*** (16.75)
Constant	-0.28* (0.20)	-0.40** (0.20)	-0.57*** (0.19)	-0.39* (0.20)	-0.68*** (0.20)	-0.87*** (0.20)	-0.73*** (0.22)	-1.75*** (0.28)	-1.92*** (0.29)	-1.71*** (0.35)	-2.18*** (0.38)	-2.16*** (0.44)	-1.80*** (0.50)
Observations	156	156	156	156	156	156	156	156	155	154	153	152	151
R ²	0.42	0.42	0.42	0.39	0.40	0.40	0.30	0.32	0.32	0.21	0.24	0.20	0.15

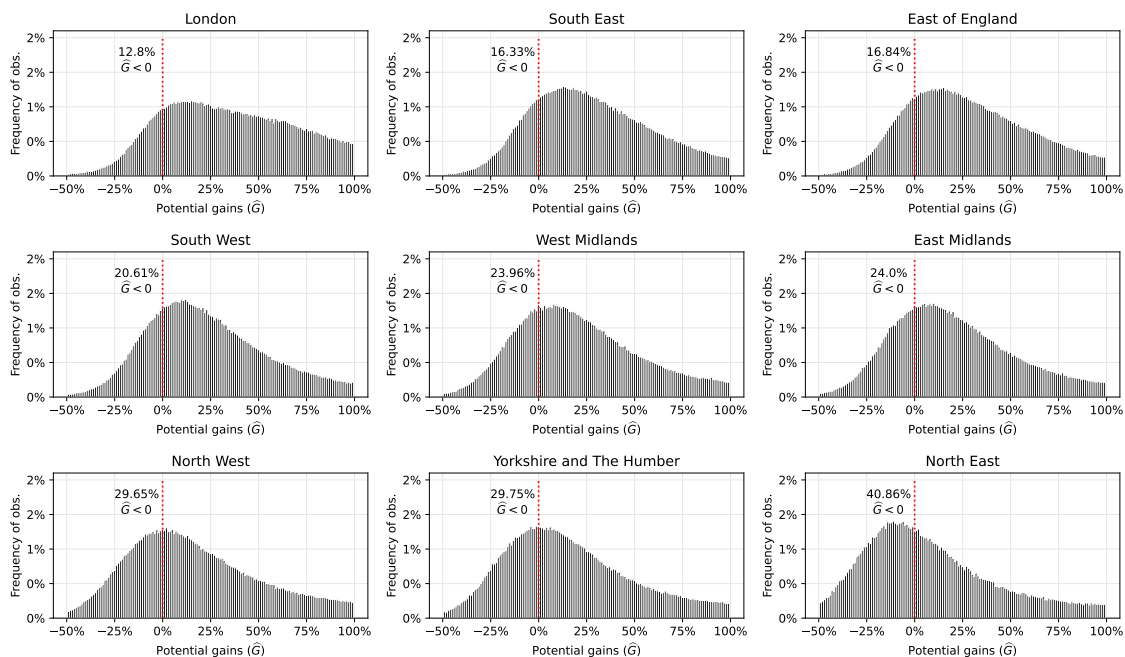
Figure A.10
 “Paper losses” and home equity

The figure reports histograms for potential home equity, i.e., the percent differences between hedonic property values and outstanding mortgage amounts. The data cover the period between 2015 and 2022, and are recorded in half-yearly snapshots as part of the PSD007 data set of the Financial Conduct Authority.

Panel A
 Full-sample distributions



Panel B
 Cross-regional heterogeneity



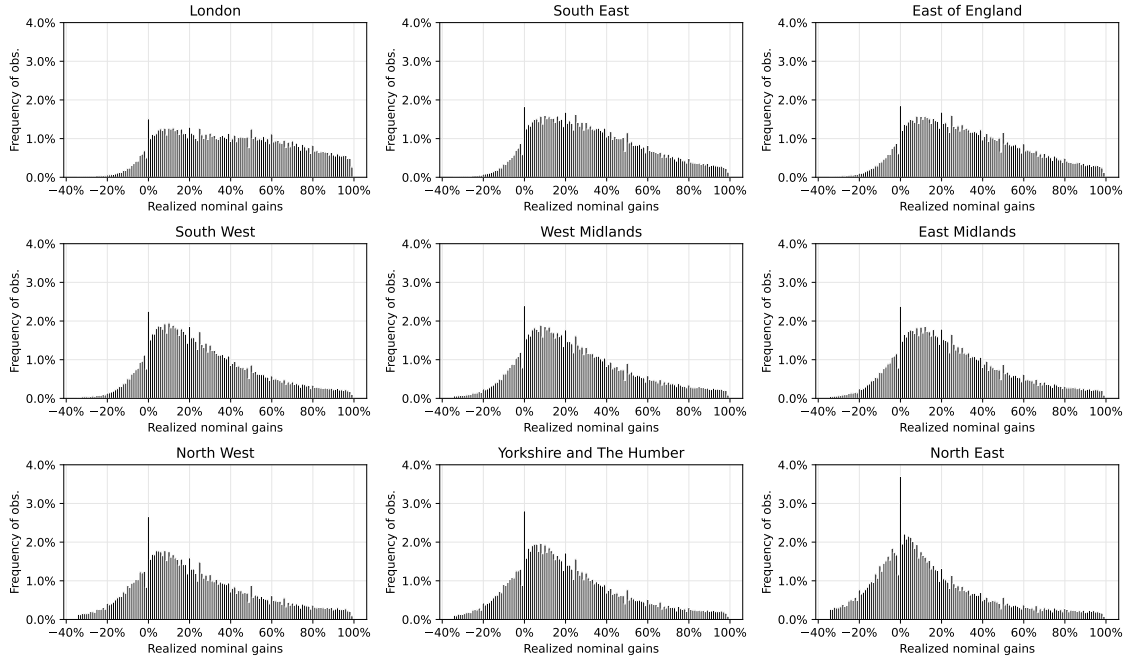
A.2 Bunching of transaction prices

Ex ante, as initially observed by [Genesove and Mayer \(2001\)](#), prospective reference-dependent and loss-averse sellers of houses with similar current market value set different list prices, depending on their original purchase prices. Sellers facing nominal losses relative to the reference point set list prices significantly higher, to “fish” for buyers willing to pay a high price, even if this implies a much longer waiting time to sell the property.

Ex post, reference-dependent and loss-averse sellers steer transaction outcomes away from realized nominal losses and towards realized gains. The resulting distribution of final transaction prices exhibits a sharp bunching of transaction prices exactly at the reference point and a significant shift in total mass away from the domain of nominal losses.

Figure A.11
Regional variation in bunching of transaction prices

The figure reports the distribution of differences between the realized price in a given sale transaction and the price for which the property has been initially purchased (the nominal realized gain), computed for nine different ITL1 regions of the U.K.



To document these patterns, for each property for which at least two transactions are recorded in the HM Land Registry data, we calculate the *ex post* nominal capital

gain as the difference between the observed re-sale transaction price and the original purchase value. Figure A.11 reports the frequency distribution of these nominal gains realized over the period 1995–2022, for binned percentage point intervals that are closed to the left and open to the right (i.e., the 0% bin contains gains in the interval between 0% and strictly below 1%), for nine ITL1 regions in England.

Consistent with prior evidence, for every one of the regions of the U.K., we find missing mass immediately below the reference point and significant bunching at exactly zero nominal gains. An additional build-up of diffuse mass shows up for low and positive nominal gains, consistent with reference dependence in the presence of optimization frictions (Anagol et al., 2022).¹

What is new relative to prior literature is the striking variation that we observe across regions—because of the different trajectories of local housing markets, there is wide variation in the levels of realized nominal gains across regions of the U.K. As our model later predicts, these different levels of “paper losses” across regions are a key policy-relevant statistic. They encode the history of decisions by market participants in these regions and determine how the market will respond to different interventions going forward.

A.3 Seller optimization: Listing premia

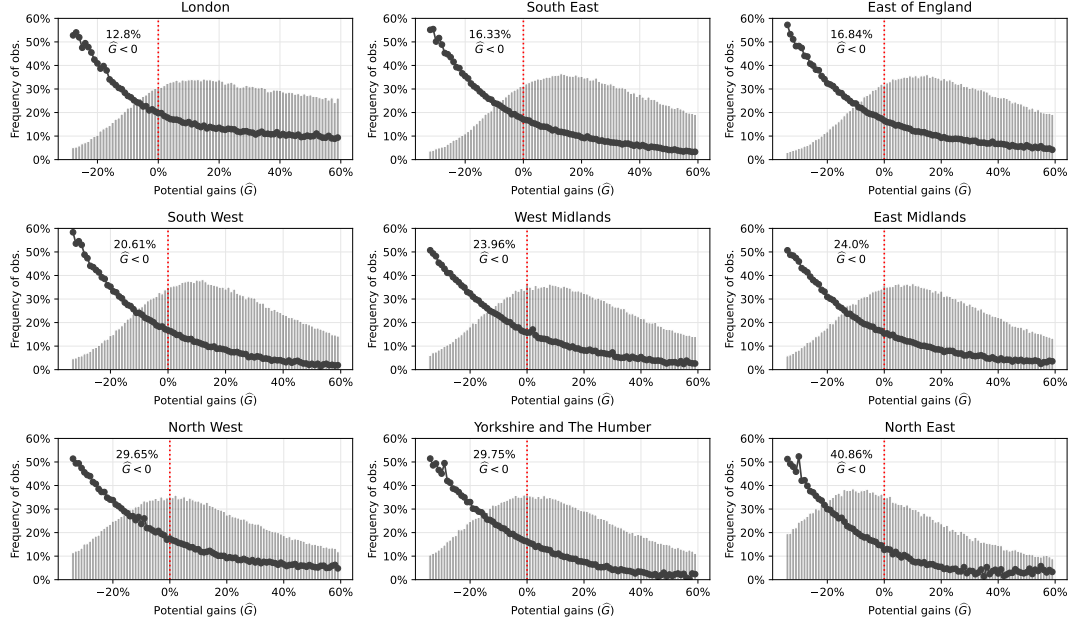
To document ex-ante reference-dependent seller behavior, we plot initial listing prices set by sellers on the online platform, i.e., the step preceding bilateral negotiations and contractual outcomes. We calculate the potential gain of a listing ($\hat{G}$) as the percent difference between the hedonic valuation of the property and its initial purchase price, and the listing premium (ℓ) as the percent difference between the listing price and the corresponding hedonic valuation.

Figure A.12 shows that the listing premium is negatively sloped with respect to potential gains and that there is a change in this slope around zero potential gains across all regions. The negative slope is consistent with reference dependence and the non-linearity with the presence of loss aversion: as sellers face nominal losses, they are more likely to engage in “fishing” behavior, i.e., to list for higher prices relative to a measure of fair market value; this strategy becomes more aggressive, the higher the magnitude of the negative potential gain.

¹This strict and diffuse bunching at zero nominal gains in all local housing markets shows up over and above any tendency of transaction prices to cluster at round numbers.

Figure A.12
“Hockey stick” pattern across regions

The figure reports average listing premia, i.e., the percent differences between the listing price and the hedonic property value at the time of listing, for different values of potential gains. We restrict the horizontal axis, for a more convenient graphical representation of effects around zero potential gains. The frequency distribution of potential gains is shown in gray shading. The numbers reported in the left upper corner of each plot indicate the share of home-owners in each ITL2 region for the nominal potential gain has a value below zero.



The *ex ante* pattern of listing premia variation across potential gains is robust across geographical locations, consistent with the observed bunching of transaction prices *ex post*. As [Stein \(1995\)](#) highlights, mortgage leverage also generates a desire to “fish” as it magnifies the effects of losses on home equity, reducing the expected down payment on the next house, thus increasing the likelihood of painful downsizing. Table [A.10](#) explores the relationship quantitatively in the full sample of listings and isolates sellers’ response to their nominal gain position from their level of home equity, attributing the bulk of the effect to the former.²

²Another way to see that the effect of losses exists over and above the effects of mortgage leverage is in Panel A of Figure [A.2](#), which reports the mean listing premium in each potential gain percentage point bin for two different samples. The first is a sample of listings for which a mortgage contract is outstanding at the time of listing (“Mortgage sample”), and the second is listings where the mortgage has either been completely repaid or the seller’s ownership is outright.

Table A.10
Listing premia: Conditional “hockey stick” patterns

The table reports estimated coefficients from the following regression specification:

$$\ell_j = \beta_1 1_{\hat{G}_j < 0} \hat{G}_j + \beta_2 1_{\hat{G}_j \geq 0} \hat{G}_j + \beta_3 1_{\hat{H}_j < 0} \hat{H}_j + \beta_4 1_{\hat{H}_j \geq 0} \hat{H}_j + \epsilon_j,$$

where j indexes an individual listing. We restrict the sample to listings which are successfully merged between the Rightmove listing data, transaction records in the Land Registry, and the Bank of England mortgage balance data. *, **, *** indicate statistical significance at the 10%, 5% and 1% level.

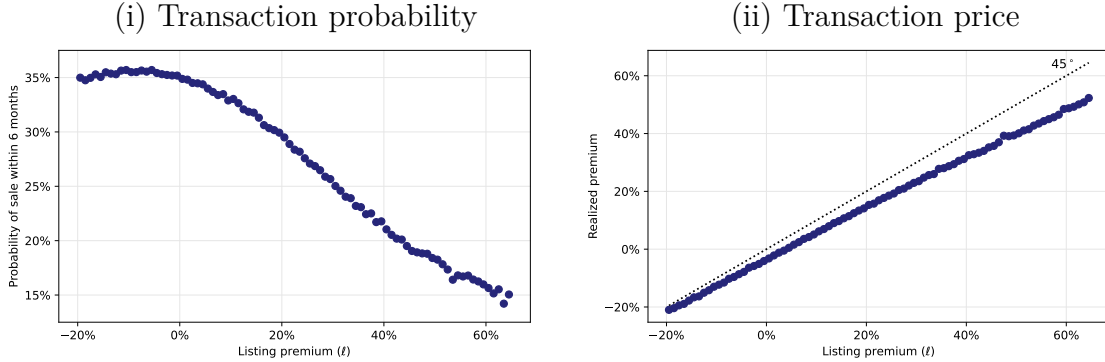
Dependent variable: Listing premium (ℓ)		
Potential gain ($\hat{G}$)	-0.48***	
	(0.005)	
$\times \hat{G} \leq 0$	-1.081***	
	(0.009)	
$\times \hat{G} > 0$	-0.357***	
	(0.005)	
Potential home equity ($\hat{H}$)	-0.434***	
	(0.007)	
$\times \hat{H} \leq 20\%$	-0.125***	
	(0.010)	
$\times \hat{H} > 20\%$	-0.098***	
	(0.008)	
No. of obs.	150,587	150,587
R ²	0.416	0.444

A.4 Buyer optimization: Matching outcomes

How do buyers respond to reference-dependent sellers’ listing strategies? Plot (i) of Figure A.13 shows the probability that a given property is sold within six months (180 days) post-listing. The average probability is 33%; but, more importantly, the probability of a quick sale varies strongly with the price premium the seller chooses when listing the property. The greater the difference between the initial asking price and the estimated hedonic value at the time of listing, the lower the probability that the sales will occur within six months. Plot (ii) of Figure A.13 illustrates how final sales price premia (sales price less hedonic value) track listing premia. There is a strong positive relationship between the two, consistent with sellers trading

Figure A.13
Search and matching outcomes

The figure reports outcomes of the search and matching process, for different levels of the sellers' chosen initial listing premium. Plot (i) shows the average probability that a given listing results in a successful sale within 6 months. Plot (ii) shows the average realized sale price, relative to the estimated hedonic value of the property, i.e., the realized premium.



off longer time on the market for eventually higher transaction prices arising from meeting buyers with higher willingness to pay. Essentially, this is a downward-sloping demand curve. Higher listing premia from aggressive “fishing” by sellers with low potential gains or low home equity reduces the likelihood of an eventual sale or, at a minimum, increases the property’s pre-sale time on the market.

A.5 Notes on the construction of the mortgage data

Column 1 of Table A.3 shows the total number of observations of mortgages across different snapshots that also report property values; column 2 shows the number of observations which report a unique combination of granular postcodes, property value, and transaction dates; column 3 shows the number of observations that report granular postcodes and borrower date of birth. The decline in the number of observations from columns 2 to 3 shows that the variable combination we use to track mortgages across snapshots (i.e., column 3) is more restrictive than the unique combination of postcode-purchase price-date of transaction in column 2. More specifically, when moving from column 2 to 3, we drop cases where a borrower has multiple liens on the same property, some of which are duplicate observations. We, therefore, rely on the sample in column 3 for all the moments computed using the mortgage data to avoid any noise generated by measurement error.

Table A.4 shows that the binned versions of the home equity variables have the same mean but a lower standard deviation than their non-binned counterparts.

We use the mortgage balances tracked over multiple snapshots to calculate an amortization schedule. Since mortgage borrowers in the U.K. may close their accounts when moving houses or refinancing, we restrict this sample to contracts with a non-zero balance in each half-yearly snapshot between 2015H2 and 2022H2. Thus, we condition on the particular postcode–date of birth combination being present in all subsequent snapshots, with a non-zero balance on the associated mortgage. In results available upon request, we find that the amortization schedule is similar across origination dates and borrower types, and does not differ greatly for mortgages with different loan-to-value ratios at origination.

A.6 Listings vs. transaction volumes

In the listings sample, we can decompose the price-volume regression to listing volumes and transaction probabilities as

$$\ln(V_t) = \underbrace{\ln(V_t/L_t)}_{\text{Probability of transaction conditional on listing}} + \underbrace{\ln(L_t)}_{\text{Listing volume}}.$$

We implement this decomposition by considering whether listings in the Rightmove listings data are transacted within 180 days of the first listing. Leveraging this linear structure we obtain $\beta = \beta_V + \beta_{VL}$, where

$$\ln(L_t) = \gamma_L + \beta_L P_{\Delta,t} + \varepsilon_{t,V}, \quad (15)$$

$$\ln(V_t/L_t) = \gamma_{VL} + \beta_{VL} P_{\Delta,t} + \varepsilon_{t,VL}, \quad (16)$$

The results for these price-volume regressions are shown in Table A.5. We observe that approximately 60% of the price-volume elasticity is explained by the intensive margin, i.e., the probability that sales listings convert to actual transactions.

We note that transactions volumes are obtained from the Land Registry transactions data. To consider the *additive* decomposition of transaction volumes to listing volumes and listings converting to transactions, we need to build the transaction volumes from the listings data. This restricts the sample to the set of transactions for which we observe a listing in the Rightmove listings data.

A.7 Loss share as sufficient statistic: Analytical derivations

Consider a partition of the (r, m) domain into J bins, each with mass N_j of home-owners. Denoting by $\bar{s}_{jt}^*$ and $\bar{l}_{jt}^*$, the binned means of $s_{it}^*(\theta)$ and $l_{it}^*(\theta)$, respectively for each $\theta \in (-\infty, \infty)$, we can now approximate the right-hand side of the above formula using the corresponding Lebesgue sum as

$$\text{Transaction Volume} \approx \sum_{j=1}^J N_j \int_{-\infty}^{\infty} \bar{s}_{jt}^*(\theta) \alpha(\bar{l}_{jt}^*(\theta)) d\Phi_{\theta}(\theta).$$

Let $\bar{\theta}_{jt}^*$ denote the binned mean of the threshold θ for listing in bin j . Using the second order approximation for $\alpha(\cdot)$, we obtain

$$\begin{aligned} \text{Transaction Volume} &\approx \alpha_o \sum_{j=1}^J N_j \int_{-\infty}^{\infty} \bar{s}_{jt}^*(\theta) d\Phi_{\theta}(\theta) \\ &\quad - \alpha_1 \sum_{j=1}^J N_j \int_{-\infty}^{\infty} \bar{s}_{jt}^*(\theta) \bar{l}_{jt}^*(\theta) d\Phi_{\theta}(\theta) \\ &\quad - \alpha_2 \sum_{j=1}^J N_j \int_{-\infty}^{\infty} \bar{s}_{jt}^*(\theta) (\bar{l}_{jt}^*(\theta))^2 d\Phi_{\theta}(\theta) \\ &\approx \alpha_o N_{St} - \sum_{j=1}^J N_{Sjt} [\alpha_1 \bar{\ell}_{jt}^* + \alpha_2 (\hat{\ell}_{jt}^*)^2], \end{aligned} \tag{17}$$

where we have denoted the number of listings in bin j by

$$N_{Sjt} \equiv [1 - \Phi(\bar{\theta}_{jt}^*)] N_j = N_j \int_{-\infty}^{\infty} \bar{s}_{jt}^*(\theta) d\Phi_{\theta}(\theta), \quad \text{and} \quad N_{St} = \sum_{j=1}^J N_{Sjt},$$

and the binned listing premium conditional on listing by

$$\bar{\ell}_{jt}^* \equiv \frac{1}{1 - \Phi(\bar{\theta}_{jt}^*)} \int_{\bar{\theta}_{jt}^*}^{\infty} \bar{l}_{jt}^*(\theta) d\Phi_{\theta}(\theta)$$

and

$$\hat{\ell}_{jt}^* \equiv \left[\frac{1}{1 - \Phi(\bar{\theta}_{jt}^*)} \int_{\bar{\theta}_{jt}^*}^{\infty} \bar{l}_{jt}^*(\theta)^2 d\Phi_{\theta}(\theta) \right]^{1/2}.$$

Equation (17) can now be used to approximate the model-implied transaction volumes with high precision as the number of bins $J \rightarrow \infty$. In particular, equation (14) now follows from considering a partition with $J = 2$, with only two bins $\{r_i > \bar{p}\}$ and $\{r_i \leq \bar{p}\}$, where $\bar{p}$ denotes the average (log) transaction price, and approximating the conditional L^2 -norm above by the L^1 -norm. In practice, in our empirical tests of this approximation we consider interaction terms of loss share and listing premium with second order terms for the listing premium.

A.8 Computational appendix

The model is solved by value function iteration on a two-dimensional grid of reference prices and mortgages. The implementation is otherwise standard, but for the mortgage grid we provide the following additional details.

We characterise $F_m(\cdot)$ as a discrete distribution by assuming a discrete grid of mortgage loan-to-value (LTV) ratios relative to the original purchase price of the property: $\tilde{m}_1 > \tilde{m}_2 = \tilde{m}_1 - \delta_m > \dots > \tilde{m}_{n_m} = \tilde{m}_1 - (n_m - 1)\delta_m$ with weights $w_1^m, \dots, w_{n_m}^m$ at issuance. The mortgage grid runs from 42.8% LTV to 95% LTV (relative to original purchase price). The 42.8% LTV lower bound is low enough for no-one in the steady-state distribution to incur the downsizing penalty at this lower bound.

Each new homeowner i , therefore, has an issuance mortgage balance drawn from $\{R_i \cdot \tilde{m}_1, \dots, R_i \cdot \tilde{m}_{n_m}\}$, $r_i = \ln R_i$. Moreover, given a draw $m_{it} = r_i + \log(\tilde{m}_{l_i})$, we assume that the mortgage balance evolves along the same grid as $m_{i,t+1} = r_i + \log(\tilde{m}_{l_{i+1}})$. In what follows, we will denote the log mortgage balance by $m_l \equiv m_l(r) = r + \log(\tilde{m}_l)$, suppressing the dependence on r .

Given the discrete grid on mortgages, for the numerical implementation, we rewrite (11) as

$$\begin{aligned}
f^{k+1}(r, m_l) &= \underbrace{f^k(r, m_{l-1})}_{\text{Existing homeowners}} \times \underbrace{\left[1 - \int_{\theta^{k,*}(r, m_{l-1})}^{\infty} \alpha^k(p^{k,*}(r, m_{l-1}, \theta)) d\Phi_{\theta}^k(\theta)\right]}_{\text{Non-sellers and failed sellers}} \\
&\quad + \underbrace{N_S^k \omega^k(r) \alpha(r) w_l^m}_{\text{New homeowners}},
\end{aligned} \tag{18}$$

for $l = 2, \dots, n_m$ and $f^{k+1}(r, m_1) = N_S^k \omega^k(r) \alpha(r) w_1^m$ for $l = 1$, and where k denotes

the k th step of the value function iteration and $p^{k,*}$ sellers' optimal price setting function in the k th step..

A.9 Model calibration

For the model estimation, we need to establish the proper target objects in the data, which include both transactions data and listings data. In the model, most of the objects are defined in terms of post-bargaining prices p that would actually occur conditional on sale. These correspond to some pre-bargaining listing prices l via $p = p(l)$. This formulation encounters the problem that in the data the natural mapping between listing prices and the conditional transaction prices is determined with respect to the *listing premium* relative to the hedonic value of the property, but in the model, where all the properties are identical, we can define the listing premium only ex post after solving for the endogenous average transaction price.

Conceptually, this is a non-issue, since all the agents form rational expectations of prices. Nevertheless, we wish to avoid this for computational reasons. Instead, we solve the issue by reformulating the homeowner problem equivalently as a maximization over conditional transaction prices p rather than l . This is possible as long as the function $p(l)$ is bijective with inverse $l(p)$, which we assume.

$$\begin{aligned}
V_t^h(r_i, m_{it}) = & u_t - \tau^h + \mathbb{E}_\theta \max \left\{ \underbrace{\max_{p_{it}} \alpha_t(l(p_{it})) [U(\cdot) + \theta_{it} + \beta \mathbb{E}_t[V_{t+1}^b]]}_{\text{Listing success}} \right. \\
& + \underbrace{(1 - \alpha_t(l(p_{it}))) \beta \mathbb{E}_t[V_{t+1}^h(\cdot)]}_{\text{Listing failure}} - \phi, \underbrace{\mathbb{E}_t[\beta V_{t+1}^h(\cdot)]}_{\text{No listing}} \left. \right\}. \tag{19}
\end{aligned}$$

The benefit of doing so is twofold. First, the seller's problem now only depends on the conditional transaction prices and not on the listing prices, which in the data should be interpreted as listing premia over hedonic values. Second, we may now define a modified demand function $\tilde{\alpha}(\cdot)$ as

$$\tilde{\alpha}(p) = \alpha(l(p)) = \chi(q) \cdot [1 - F(\varepsilon^*(p))].$$

This depends on the conditional transaction price p instead of the listing price l , indeed confirming that listing prices do not enter the seller's problem. Moreover, for the estimation, we can now adjust the demand function in the data by the function $l(\cdot)$ to match the data objects with those arising from the model.

More precisely, defining the listing premium as $\ell = l - \widehat{p}$, where $\widehat{p}$ is the hedonic value of the property, corresponding to the average transaction price in the model, both functions $\alpha(\ell)$ and $\ell(p - \widehat{p})$ are observable in the data. Therefore, for each price p , we can produce $\widetilde{\alpha}(p - \widehat{p})$ in the data, which can then be targeted in the model estimation. In practice, we assume a linear form for the listing premium as function of the realized premium by estimating

$$\ell_{it} = \beta_0 + \beta_1(p_{it} - \widehat{p}) + \varepsilon_{it},$$

and set the targets in terms of data by scaling the model implied moments correspondingly. For any given model-implied value of α_{it} associated with some post-bargaining listing price p_{it} , we match that against a listing premium ℓ_{it} using the above empirical model and target α_{it} from the model against $\alpha(\ell_{it})$ in the data.

Note that each estimation step solves for the model for some parameter values, and after solving the model for any such set of parameters, we may produce $p - \widehat{p}$ also from the model for targeting the shape of $\widetilde{\alpha}$ to match that of the data.

Secondly, we also need to adjust the sellers' price setting rule. In the data, we observe the listing premium as a function of the potential gain $\ell(\widehat{p} - r)$. On the other hand, for each triplet (r, m, θ) the model gives an optimal post-bargaining price $p^*(r, m, \theta)$ set by the seller, which can be mapped to some listing premium $\ell(p^*)$ using the above empirical model. In the estimation, we then target the shape of $\ell(p^*(\widehat{p} - r))$ to match $\ell(\widehat{p} - r)$ in the data. In particular, we will average over the values of m and θ when producing the model implied price setting rule $p^*(\widehat{p} - r)$. Assuming that $\ell(\cdot)$ is linear, this averaging does not bias the estimation procedure.

Finally, when working with the conditional post-bargaining transaction prices p rather than the listing prices, we should note that the listing price distribution $\widetilde{\Omega}(p) = \Omega(l(p))$ is also modified via the change of variables. In the estimation procedure, we target moments of the listing price distribution, so we have to account for this change of variables:

$$\widetilde{\omega}(p) dp = \omega(l) \frac{dl}{l'(p)}.$$

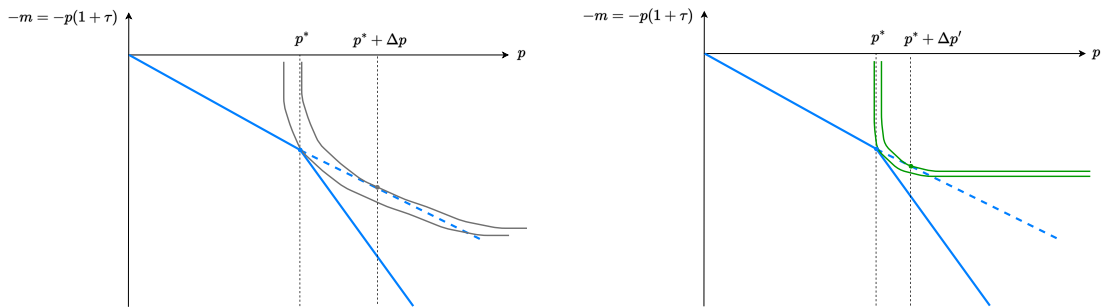
Assuming the estimated linear structure above, this implies that $\mu(\ell) = \beta_0 + \beta_1(\mu(p) - \widehat{p})$ and $\sigma(\ell) = \beta_1\sigma(p - \widehat{p})$, which are the transformations that will be used for the model implied moments in our moment matching exercise.

A.10 Theoretical intuition: Bunching at tax threshold levels

To understand the quantification of bunching effects at tax kinks and notches, define $m = p(1 + \tau)$ as the total cash outflow of the household, for a property valued at price p and a tax rate τ . shows that buyers in the interval $[p^*, p^* + \Delta p]$ bunch at p^* .

With a steeper indifference curve, the bunching mass decreases. In this context, the only decision variable of the buyer in relation to a particular property/seller is the timing of the purchase. In the labor market, the worker adjusts the number of hours to determine final income. Their trade-off is driven by the individual consumption-leisure choice. Here, the situation is similar: The buyer can choose to either buy immediately for a high price, or buy later for a lower one, if the seller is not able to sell the property in the meantime. In the labor market, a bunching worker works less; here, a bunching buyer waits longer, or threatens to do so. The magnitude of the trade-off is determined by the seller's urgency, i.e., the degree to which they are likely to decrease the price as time goes by. Note that, in contrast to a worker, a buyer also has the option to walk away from the current property, and search harder in the market. Reference dependence affects the sellers' willingness to negotiate, i.e., a seller in the loss domain is less likely to decrease the price as the time-on-the-market decreases. As the buyer's indifference curve is steeper, this leads to a lower magnitude of bunching.

Figure A.14
Illustration of bunching at tax kink points



A.11 Housing taxation in the UK

Housing consumption in the UK is subject to two taxes: a stamp duty paid by the owner at the time of buying the property, and a property tax paid monthly to the

local government by the occupant (owner-occupier or renter). Below we provide an overview of the stamp duty in the UK.

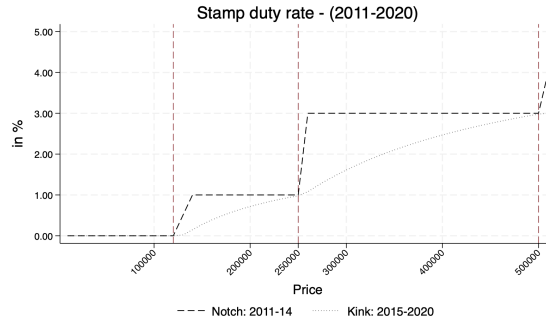
Stamp duty transaction tax is imposed on all housing transactions in the UK, and paid by the buyer. There was a substantial change in December 2014 in the way stamp duty was calculated on the transaction price. As shown in panel (A) of Figure [A.15](#), the average stamp duty paid was a notched schedule of the transaction price, with the tax paid jumping at discrete thresholds such as £125,000 and £250,000. Since Dec 2015, the average tax rate follows a more gradual profile, with kinks at the same thresholds and subsequent tax reform focusing on the slope of the tax rate between the thresholds.

Panel (B) of Figure [A.15](#) shows a large effect of the tax reform on bunching of transaction prices at the tax thresholds. We describe excess bunching mass at a price band (of size £5000) as the difference in the actual mass (based on transaction price) vs the expected mass (based on hedonic price) for the price band in a given period. The two figures in panel (B) show that the switch in the stamp duty from a notched to a kink schedule has a marked impact on bunching at the £250,000 price threshold.

Figure A.15
Transaction tax schedule and bunching

Panel A plots the average stamp duty rate paid for different levels of housing transaction prices in the UK during 2011-2014 (notch schedule) and 2015-2020 (kink schedule). The average rate paid changes at threshold prices indicated by the dashed vertical red line. Panel B shows histograms of all transactions in 2013 (under notch tax schedule) and 2016 (under kink tax schedule) by the hedonic price and the transaction mass. The difference in the mass by transaction and hedonic price for any given price band gives the bunching mass for the price level.

Panel A
Transaction tax rates for different prices (2011-2020)



Panel B
Histograms of transaction and hedonic prices in 2013 (notch schedule) and 2016 (kink schedule)

